



Clinical Decision Support Tools for Emergency Care Providers

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ABOUT THE CLINICAL DECISION SUPPORT TOOLS

The development of this collection of clinical decision-making tools was commissioned and funded by The Professional Board for Emergency Care (PBEC) of the Health Professions Council of South Africa. The tools are designed to support clinical decision-making at the emergency out-of-hospital point of care. Patients and their clinical presentations are always unique and will differ. The tools serve as an adjunct to clinical assessment and general clinical decision-making. There is an assumption that practitioners are skilled and knowledgeable within their scopes of practice.

Existing tools are referred to from the Resuscitation Council of Southern Africa (RCSA) website – https://resus.co.za/subpages/RCSA_Information/Resources/Algorithms.html

The tools are based on reference resources like the Clinical Practice Guidelines (2018) as well as the latest evidence from synthesised databases. The development process of the tools has been overseen by a Guideline Development Advisory board and each tool has been reviewed by several national experts.

It is the practitioner's responsibility to maintain their continuous professional development and familiarise themselves with these resources.

HOW TO USE THE TOOLS

1. Begin every assessment with a primary survey to identify and manage immediately life-threatening problems. Then, depending on the patient's presenting symptoms, navigate to the relevant tool using the index/contents at the bottom of the primary survey.
2. Follow the pathway to assess and manage the patient. When encountering a double-sided arrow (↔), consult the relevant page for additional management, but return to complete the assessment and management on the original page.
3. Find quick reference ranges for heart rate, respiratory rate, blood pressure and level of consciousness on the inside of the back cover.
4. Use the medication dosing charts at the back of the guide to calculate doses for children.
5. Look at the tag at the top right corner of each tool to determine whether the content pertains to adults only, children only or both adults and children.
Note: an adult is defined as ≥ 13 years (SA Hospital defn) – use clinical judgment to decide whether to use child, weight-based calculations for the older child and adolescents.
6. Use the colour-coded text and symbols, shown in the table below, to determine whether you are authorised to perform certain tasks and administer certain medications according to your scope of practice:



Follow the guidance in the tools according to your scope of practice:		
Perform the tasks and administer the medications appearing in:	Black text with no symbols	If you are a BAA, AEA, ECA, ECT, ANT or ECP.
	Green text with stars on either side	Only if you are an AEA, ECA, ECT, ANT or ECP. (Note: not all *green* meds are on scope for AEA, ECA – review in Medicines Appendix)
	^Orange text with caps on either side^	Only if you are an ANT or ECP
	~Purple text with curvy lines on either side~	Only if you are an ECP.

ABBREVIATIONS

Abbreviation	Meaning
AVPU	Alert, Voice, Pain, Unresponsive Scale
BP	Blood pressure
BVM	Bag valve mask
CPAP	Continuous positive airway pressure
CPR	Cardiopulmonary resuscitation
CRT	Capillary refill time
C-spine	Cervical spine
DBP	Diastolic blood pressure
ECG	Electrocardiogram
ETCO ₂	End tidal carbon dioxide
ETT	Endotracheal tube
FiO ₂	Fraction of inspired oxygen %
GCS	Glasgow coma scale
HR	Heart rate
IBW	Ideal body weight
IM	Intramuscular
IN	Intranasal
IO	Intra-osseous
IV	Intravenous

Abbreviation	Meaning
MAP	Mean arterial pressure (= DBP + 1/3 SBP)
MDI	Metered dose inhaler
NASG	Non-pneumatic anti-shock garment
NGT	Nasogastric tube
NPA	Nasopharyngeal airway
OPA	Oropharyngeal airway
ORS	Oral rehydration fluid
PEEP	Positive end expiratory pressure
PO	Per os (by mouth)
PPE	Personal protective equipment
PPH	Post partum haemorrhage
RCSA	Resuscitation Council of South Africa algorithm
RR	Respiratory rate
RSI	Rapid sequence intubation
SAPS	South African Police Services
Sats	Saturation %
SGA	Supraglottic airway
SPB	Systolic blood pressure
TBSA	Total Burn Surface Area

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We would like to acknowledge the paediatric medication tables and quick reference tools have been adapted from those in PACK Child Primary Care Guide for the Child 2023 Western Cape Edition, Department of Health, Western Cape Government, South Africa and The Knowledge Translation Unit at The Health Foundation of South Africa.

PRIMARY SURVEY

Use this approach for every patient you manage.

Check for and address hazards before approaching a patient.

- Evaluate scene safety.
- Ensure wearing correct PPE.
- If needed, call for additional resources.
- On approaching patient, check whether the patient notices you and if you see any life-threatening bleeding.

Rapidly assess circulation, airway and breathing to identify life-threatening problems

1. CIRCULATION

Is there life-threatening bleeding?

Yes

No

Apply direct pressure to site to stop the bleeding.

Bleeding uncontrolled

- Perform wound packing with haemostatic gauze.
- If bleeding still severe, apply tourniquet above injury.

Is the patient conscious?

No

Yes

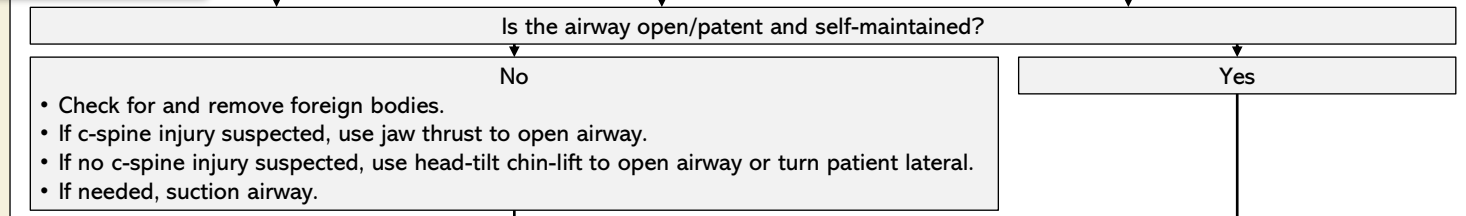
Check carotid or femoral pulse?

No pulse present

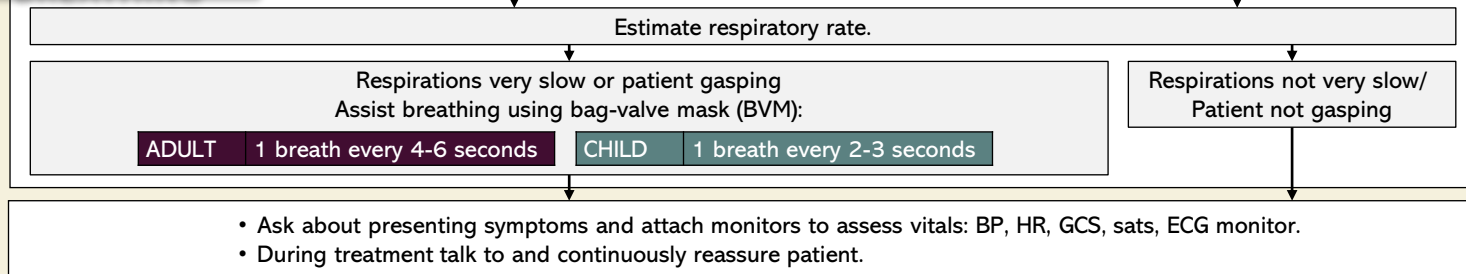
Pulse present

Start CPR
follow ↔ **RCSA: ADVANCED CARDIAC**
ARREST

2. AIRWAY



3. BREATHING



Continue to assess and manage further according to presenting symptoms and signs and age group:

All age groups

- Aggression or agitation ↔pg46
- Airway management ↔pg10
- Burns ↔pg34
- Drowning ↔pg32
- Patient in pain ↔pg58
- Poisoning ↔pg54
- Seizures/ Fits ↔pg44
- Spinal injury ↔pg38

Adult patient (≥13 years)

- Badly injured patient ↔pg28
- Breathing difficulty (adult) ↔pg20
- Chest pain ↔pg52
- Decreased consciousness ↔pg42
- Head injury ↔pg40
- Mechanical ventilation ↔pg18
- Palliative end-of-life care ↔pg66
- Shock ↔pg8
- Stroke ↔pg50

Child patient (<13 years)

- Breathing difficulty (child) ↔pg22
- Child with diarrhoea ↔pg24
- Critically ill child ↔pg26

Appendices:

- Drug Appendix ↔pg68
- Infusion guides ↔pg78
- Formulary ↔pg74
- Quick Ref Chart ↔pg79

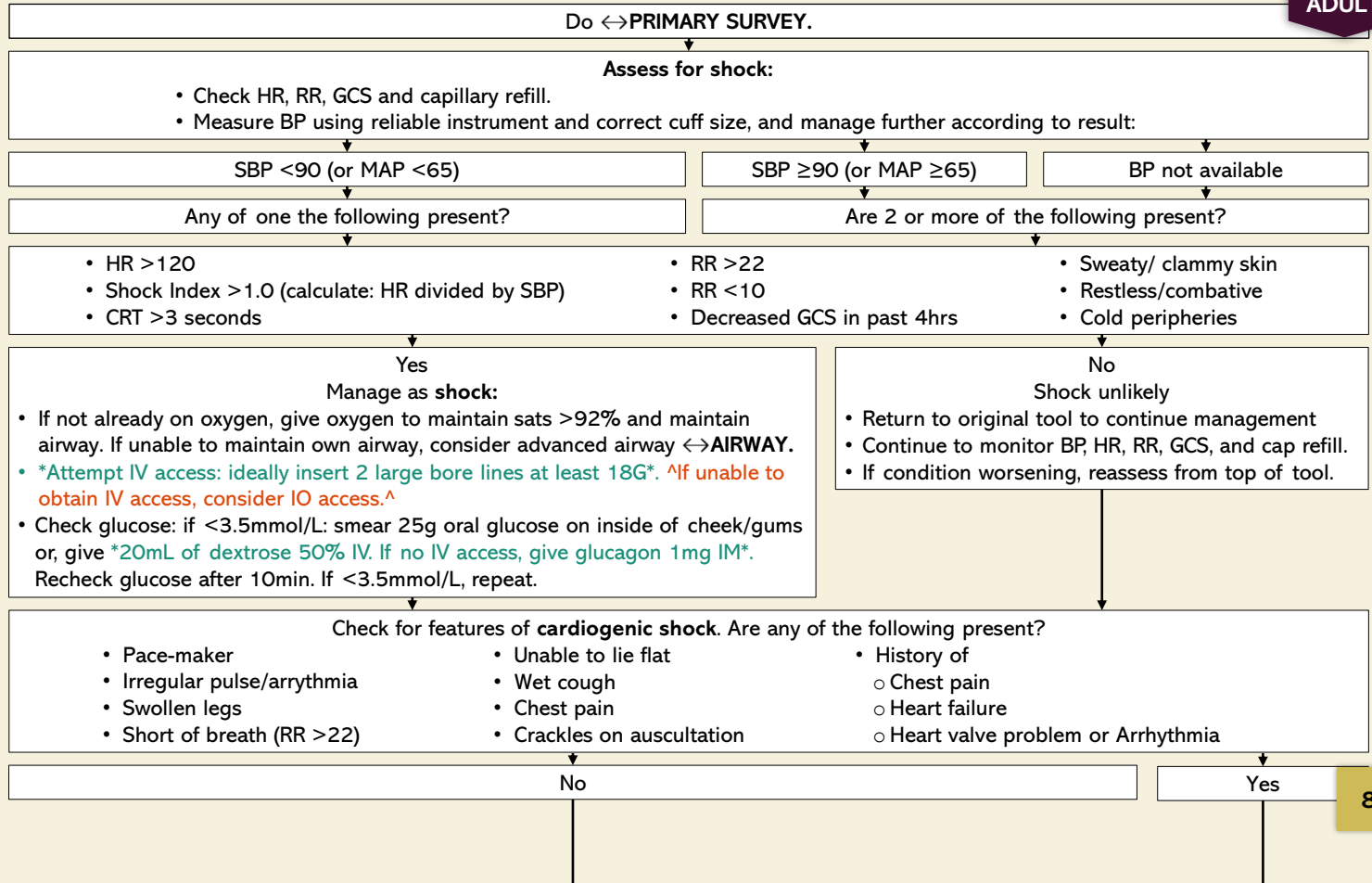
If pregnant:

- Seizures/hypertension in pregnancy (eclampsia) ↔pg62
- Vaginal bleeding after childbirth ↔pg64

SHOCK - ADULT



Consider shock and use this approach when directed from other tools, and in any critically ill or injured patient ≥ 13 years old.



Cardiogenic shock unlikely

Lie patient down and ***give crystalloid fluid 250-500mL IV over 5-10min***.

Check for features of **active bleeding**. Are any of following present?

- Obvious blood loss
- Pulsating or growing abdominal distention
- Vomiting or coughing blood
- Rectal bleeding
- Deterioration after injury: internal bleeding likely

No

If SBP still <90 (MAP <65); HR >120, RR <9, RR >20, CRT>3secs or decreased GCS:

- ***Repeat crystalloid fluid 250-500mL IV bolus over 5-10min until vitals improving.**
- **If signs of fluid overload develop (new breathlessness or new crackles on auscultation), stop fluids.***

Yes: active bleeding

- If SBP still <90, give ***crystalloid fluid 100–250mL IV and recheck BP in 10min***: repeat 100-250mL fluid boluses every 10min only until SBP ≥90 or MAP ≥65.
- **^Give tranexamic acid 2g IV over 10min.^**
- Expedite transfer.

Continue to manage likely underlying cause of shock at same time:

↔**BADLY INJURED**

↔**BURNS**

↔**POISONING**

↔**CHEST PAIN**

↔**BREATHING DIFFICULTY**

If anaphylaxis or arrhythmia, ↔**RCSA ANAPHYLAXIS** or ↔**RCSA ARRHYTHMIA**.

- If patient deteriorates despite above management, call for medical advice.
 - **^Consider adrenaline infusion.**
 - **Mix adrenaline 1mg in 50mL normal saline or 4mg in 200mL (= 20mcg/mL).**
 - **Use infusion pump or rate control device to give via peripheral line.**
 - **Avoid giving with other drugs.**
 - **Start at 5mL/hrs (=1.6mcg/min) with target dose of adrenaline in range 6-30mL/hrs (2-10mcg/min).**
- **Titrate to effect: if needed, increase by 1mL/hr every 5min until BP and HR increase.**
- **When SBP ≥90 systolic (or MAP ≥65), keep rate constant.^**

Transport to closest emergency centre.

- Keep warm/ avoid heat loss according to environment.
- Monitor and record BP, HR, RR, sats, GCS every 10min. If on adrenaline infusion, check vitals every 5min: if deteriorating, repeat primary survey ↔**PRIMARY SURVEY**.
- Manage underlying conditions.
- Assess and manage pain - cautiously as opioids may drop BP ↔**PAIN**.
- Place patient in position of comfort.
- Continue IV fluids as indicated by clinical condition.

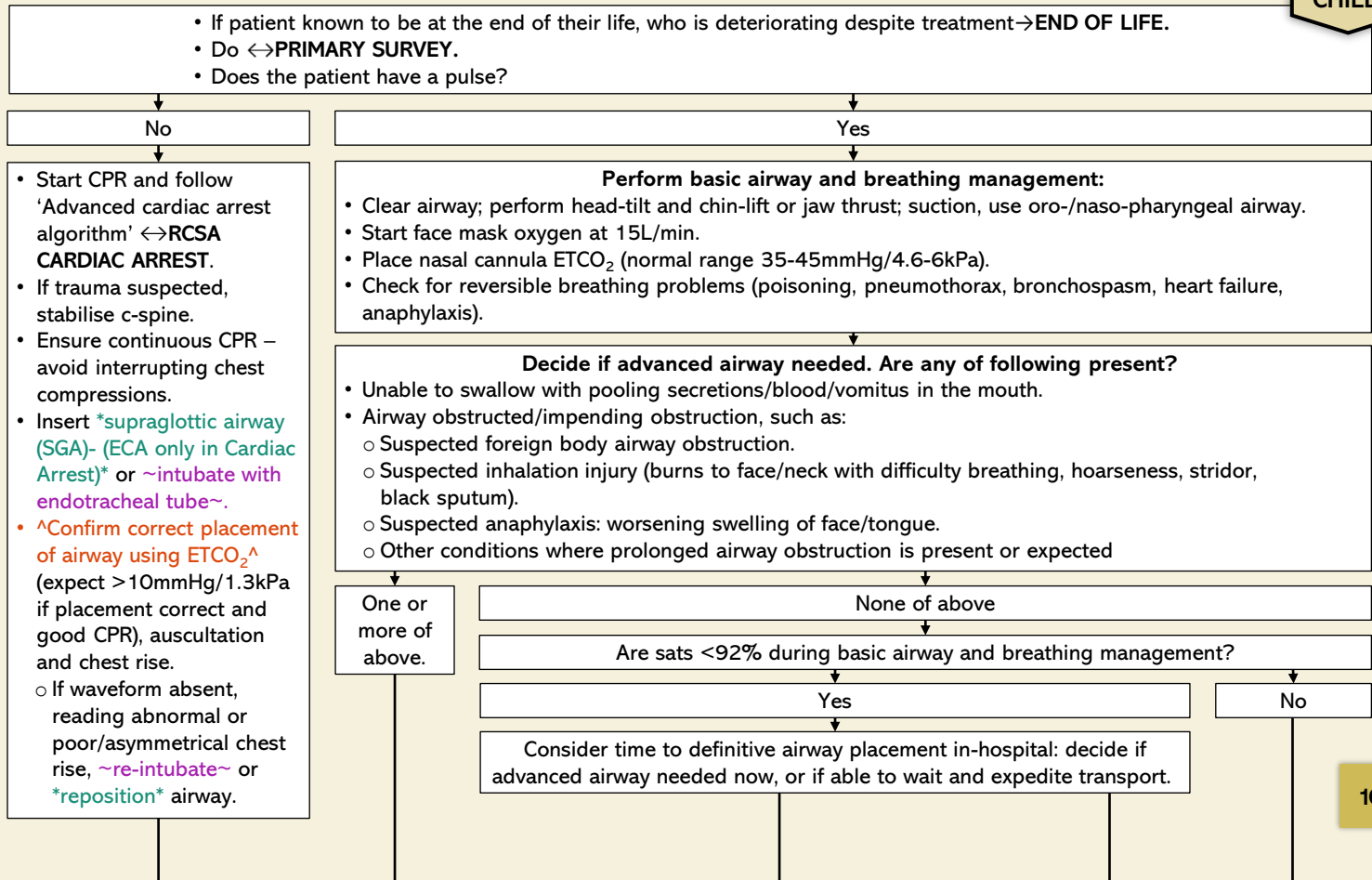
Cardiogenic shock

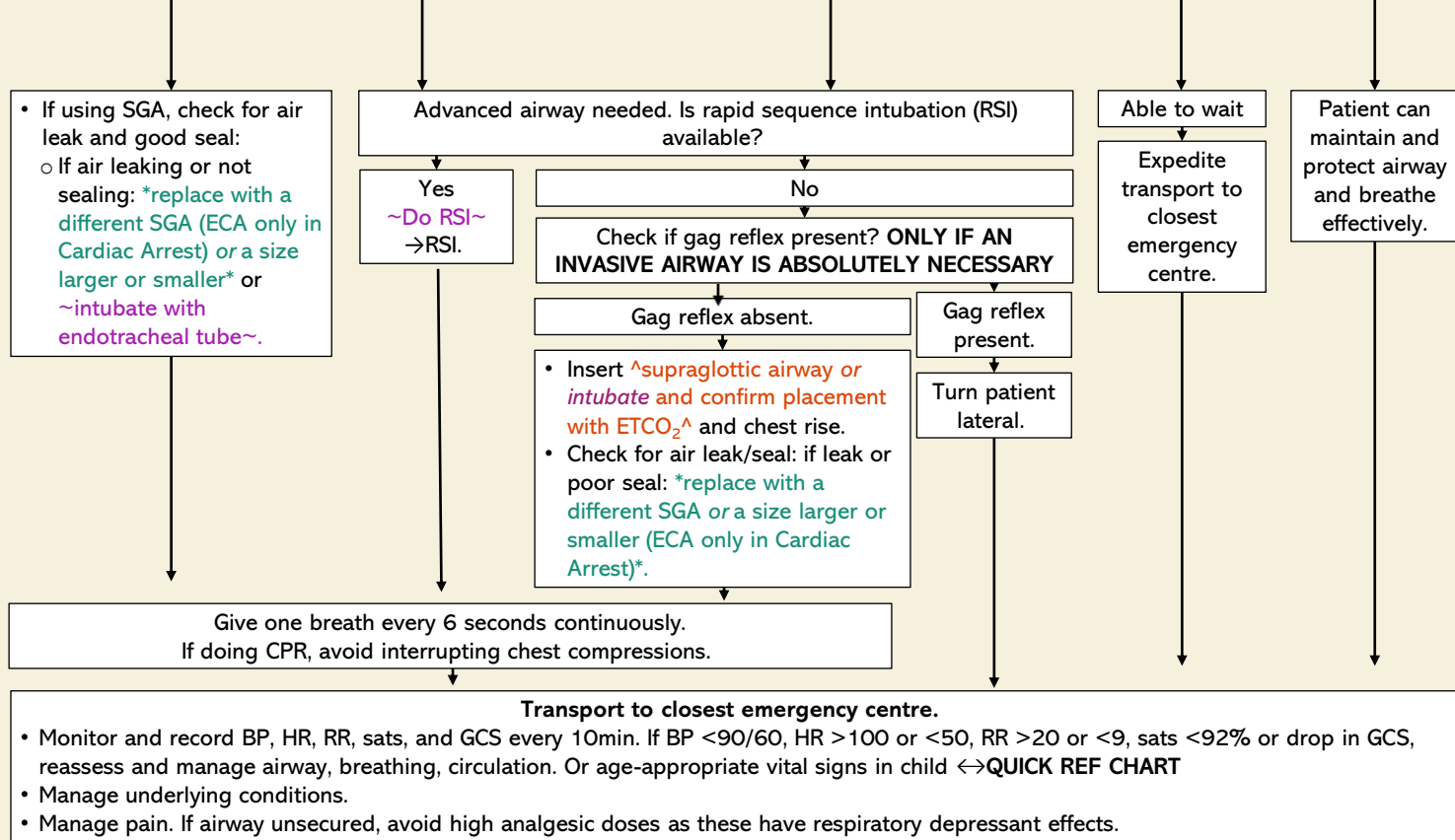
Restrict IV fluids: give crystalloid fluid at 5-10mL/hr only to keep vein open.

AIRWAY MANAGEMENT



Use this tool for the patient who is not maintaining their own airway adequately





RAPID SEQUENCE INTUBATION (RSI) PROCEDURE for Emergency Care Practitioners



ECPs to use this tool for the patient who needs a rapid sequence intubation procedure performed.

Consider the following before proceeding to RSI (as well as relevant airway assessments):

- If not yet done, treat patient's underlying condition as per relevant page. Manage shock early to avoid post-intubation hypotension.
- Assess the need to use rapid sequence intubation $\leftrightarrow$ **AIRWAY**.
- Assess procedure-related risks (negative effects on patient's haemodynamics, intracranial pressure).
- Consider other factors (time delay on scene, distance to hospital, resources available at the receiving facility).

If decision is to proceed with RSI, follow steps 1-9 to prepare.

1. Suction

Turn suction on and place hard-tip catheter under patients' right shoulder for easy access.

2. Prepare oxygen

Ensure two oxygen cylinders available.

3. Pre-oxygenate

- If normal respiratory effort, give nasal cannula oxygen 15L/min and non-rebreather mask oxygen 15L/min **OR** give CPAP at 5-10cmH₂O.
- If reduced respiratory effort, give nasal cannula 15L/min and bag-valve mask ventilations (15L/min), with PEEP valve set to 5cmH₂O, 1 breath every 4-6 seconds.

4. Airway devices

- **Endotracheal tube (with HME Filter):** choose correct endotracheal tube size and ensure two spare tubes available, a size smaller and a size larger.
 - Check cuff integrity, ensure access to 10mL syringe for cuff inflation, lubricant, and tape for securing the endotracheal tube.
- **Laryngoscope:** check that the light works, that correct blade sizes available (Macintosh or Miller blade) and ensure spare handle available.
- **Introducer:** ensure gum elastic bougie and stylet available.
- **Supraglottic device:** ensure supraglottic devices available in ideal size with two spares, a size smaller and larger.
- **Surgical/needle cricothyroidotomy:** ensure access to scalpel, paediatric bougie, 5.5 endotracheal tube/14-gauge catheter-over-needle.

5. Monitoring equipment:

- **Cardiac monitoring:** attach ECG monitoring to patient and ensure screen in view of intubating practitioner and audio settings on.
- **ETCO₂:** ensure ETCO₂ monitoring available.
- **BP and pulse oximetry:** attach BP cuff to arm without IV access and move pulse oximetry probe to arm without BP cuff.

6. Medication

Draw up analgesia, induction and paralytic agents according to patient presentation. Label syringes ready for administration.

IV analgesia:

- Fentanyl:
 - adult: 1-3mcg/kg
 - child: 1-3mcg/kg

AND

IV induction:

Consider halving the dose for hypotensive patients.

- Ketamine:
 - adult: 2mg/kg
 - child: 1-2mg/kg
- Etomidate:
 - Adult: 0.3mg/kg
 - Child: 0.3mg/kg

AND

IV paralytic:

- Rocuronium:
 - adult: 1.2mg/kg
 - child: 1.2mg/kg
- Succinylcholine:
 - adult: 2mg/kg
 - child: 1-2mg/kg

7. IV access:

Establish reliable, flowing IV-line access. Ensure no obstruction proximal to IV site.

8. Patient positioning

- **C-spine considerations:** if c-collar in place: remove. Maintain neutral position: assistant positioned next to the patient and place hands on either side of the patient's head to maintain position. Assistant to avoid applying traction.
- **Normal:** position patient supine, head slightly elevated into sniffing position, with earlobe slightly above sternal notch. Position patient's face parallel with ceiling.
- **Pregnant or obese:** place blankets/pillows beneath upper body to elevate it until sternal notch and external auditory meatus (ear hole) are aligned. If pregnant, place wedge under right flank to tilt patient towards their left.
- **Child/infant:** position child's head in a neutral position using a sheet/roll under the shoulders.

9. Pre-treatment

When time allows; consider pre-treating to lessen negative effects of induction and laryngoscopy.

- **If hypotensive:** after appropriate use of fluids (e.g., septic/anaphylactic/neurogenic/haemorrhagic shock), consider:
 - adrenaline: 5-20mcg bolus (inject 1mg of 1:1000 adrenaline into 200mL sodium chloride 0.9% (5mcg/mL solution) and draw 4mL into syringe. Give 1mL of mixture IV at a time, titrating to effect).

Continue Step 9 on next page

- **If suspected raised intracranial pressure** (e.g. traumatic brain injury, stroke) or **cardiovascular risk** (e.g. acute coronary syndrome, aortic dissection):
 - fentanyl (if not already given): give 3mcg/kg IV over 30 to 60 seconds. Avoid giving in the 3min before the induction medication is administered.
- **If agitated:** (e.g. due to hypoxia, traumatic brain injury) and unable to adequately pre-oxygenate/monitor, give:
 - ketamine: 0.5-1mg/kg IV
 - fentanyl (if not already given): 1-3mcg/kg IV
- **If infant <3 months old:** consider giving:
 - atropine: 0.02mg/kg (min 0.1mg, max 0.4mg)

Place airway

- Give induction and paralytic agents and flush. Wait at least 30 seconds. Ensure patient is paralysed.
- Perform laryngoscopy. Visualise vocal cords, insert endotracheal tube (with/without gum elastic bougie) through vocal cords.
- Can endotracheal tube be visualised passing through vocal cords?

Yes

No

- Re-oxygenate with BVM at 15L/min.
- Re-evaluate and consider changing person intubating/introducer/patient position/ETT size/laryngoscope blade size.
- Only if sats >92%, perform 2nd attempt at laryngoscopy.
- Can endotracheal tube be visualised passing through vocal cords?

Yes

No

- Re-oxygenate with bag-valve mask at 15L/min.
- Place supraglottic device.
- Are sats improving and is ETCO₂ reading visible?

Yes

No

Consider re-attempting laryngoscopy or keep supraglottic in place.

Do needle or surgical cricothyroidotomy

Confirm correct airway placement

- Look at ETCO₂ reading/waveform: this should be 35-45mmHg/4.6-6kPa.
- Auscultate over left axilla, then epigastrium, then right axilla to confirm air entry in lungs (not stomach).
- Look for good and equal chest rise.
- Assess depth of ETT: typically 22cm at teeth for adult male; 20cm at teeth for adult female, child ↔ **QUICK REF CHART**
- Secure the ETT, place blanket roll/headblocks. Then recheck ETT depth.

Ventilate mechanically

- If normal lungs, initial settings are:
 - Tidal volume: 6-8mL/kg (ideal body weight)
 - Respiratory rate: 12 - 20 breaths per min (if pre-intubation rate was significantly faster, set it at that rate – adjust according to ETCO₂ reading)
 - FiO₂: if sats 100%, titrate down to 0.4 where relevant
 - PEEP: 5 – 10mmHg (consider higher levels of PEEP in lung protective ventilation)
 - Inspiratory flow rate: 60-80L/min
 - → **VENTILATION**

Give post-intubation care

- Reassess HR, BP, sats and ETCO₂.
- Raise the head of the bed slightly.
- Keep patient warm.
- Administer analgesia and sedation infusion (see doses below).
- Consider inserting nasogastric tube and placing on free drainage.
- Assess regularly for ETT displacement, ETT obstruction, pneumothorax, equipment failure and secretions.

- Continue to manage underlying conditions.
- Transport to closest emergency centre.
- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP <90/60, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation. Or age-appropriate vital signs in child ↔ **QUICK REF CHART**.
- Manage pain ↔ **PAIN**.

Post-intubation analgesia and sedation dosing

If haemodynamically stable, use morphine and midazolam IV (if haemodynamically unstable – use reduced dosages)

	Drug and dose		Mixture and rate for long distance transportation	
Adult	Midazolam	• Loading dose: 0.05 to 0.1mg/kg • Maintenance: 0.06-0.24mg/kg/hr	45mg midazolam + sodium chloride 0.9% in 45mL total volume	Run at 5-15mL/hr
	Morphine	• Loading dose: 0.05-0.1mg/kg • Maintenance: 0.02-0.08mg/kg/hr	20mg morphine + sodium chloride 0.9% in 50mL total volume	Run at 5-15mL/hr
Child	Midazolam	• Loading dose: 0.05 to 0.2mg/kg • Maintenance: 0.05-0.2mg/kg/hr	0.5mg/kg midazolam + sodium chloride 0.9% into 50mL total volume	Run at 5-20mL/hr
	Morphine	• Loading dose: 0.05-0.3mg/kg • Maintenance: 0.02-0.08mg/kg/hr	0.25mg/kg morphine + sodium chloride 0.9% into 50mL total volume	Run at 4-16mL/hr

OR

If haemodynamically unstable (in the absence of catecholamine depletion), use ketamine and midazolam IV

	Drug and dose		Mixture and rate for long distance transportation	
Adult	Ketamine	• Maintenance: 0.25-0.5mg/kg/hr	200mg ketamine + sodium chloride 0.9% in 50mL total volume	Run at 5-10mL/hr
	Midazolam	• Maintenance: 0.05mg/kg/hr	15mg midazolam + sodium chloride 0.9% in 50mL total volume	Run at 15mL/hr
Child	Ketamine	• Maintenance: 1-2mg/kg/hr	6mg/kg ketamine + sodium chloride 0.9% in 50mL total volume	Run at 8-16mL/hr
	Midazolam	• Maintenance: 0.05mg/kg/hr	0.25mg/kg midazolam + sodium chloride 0.9% in 50mL total volume	Run at 10mL/hr

This page has intentionally been left blank to help with the user-friendliness of the guide.

INITIAL SETTINGS FOR MECHANICAL VENTILATION – ADULT



The tool is for both ANTs and ECPs to use this tool to guide initial settings for mechanical ventilation in patients who have been intubated

Check the following to determine likely underlying pathology to guide ventilator settings:

- Obtain history from family/friend: ask if known asthma/COPD/emphysema/heart failure/recent pneumonia/chest trauma.
- Listen for crackles or wheeze on auscultation.
- Match findings to most likely lung pattern below and adjust ventilator settings accordingly.
- These are initial settings and will need to be optimised. Consult with medical command or receiving facility early.

- Clear lung fields
- No chest trauma
- Equal air entry

No lung disease
(normal lungs) likely

Use the following settings:

Setting:	Pressure-cycled SIMV ^{1,2}
Insp. pressure:	15-25cmH ₂ O
Tidal volume:	6-8mL/kg, IBW (up to 10mL/kg)
Ventilator rate:	12-15bpm
I:E ratio:	1:2 or 1:3
PEEP:	5cmH ₂ O
FiO ₂ :	1.0 initially, but decrease to target sats of 94-98%
Peak pressure limit:	<25-30cmH ₂ O

- Wheezing on auscultation
- Known with asthma/COPD/bronchitis/emphysema
- No breath sounds/silent chest

Obstructive lung disease likely

Use the following settings:

Setting:	Volume-cycled SIMV ¹
Tidal volume:	6-8mL/kg, IBW
Ventilator rate:	8-10 bpm
I:E ratio:	1:3 or 1:4
PEEP:	3-5cmH ₂ O
FiO ₂ :	1.0 initially but decrease to target sats of 94-98%
Peak pressure limit:	35-50cmH ₂ O

- Monitor Pplat, target <25-30cmH₂O.
- Monitor auto-PEEP.
- Manage wheeze ↔ **DIFFICULTY BREATHING.**
- Hypercapnoea permitted, don't compensate with RR.

- Crackles on auscultation
- Pulmonary oedema e.g. heart failure
- Pulmonary contusion e.g. chest trauma
- Pneumonia (cough, fever, tachycardia)
- ARDS e.g. septic

Restrictive lung disease likely

Use the following settings:

Setting:	Pressure-cycled SIMV ^{1,2}
Insp. pressure:	15-25cmH ₂ O
Tidal volume:	4-6mL/kg, IBW
Ventilator rate:	15-25bpm
I:E ratio:	1:1.5-1:2
PEEP:	5-15cmH ₂ O
FiO ₂ :	1.0 initially but decrease to target sats of 90-94%
Peak pressure limit:	30-40cmH ₂ O

- Monitor Pplat, target <30cmH₂O.
- Hypercapnoea permitted, compensate with RR.

1. If only CMV or A/C is available, optimise sedation and analgesia and continue ventilation strategy. If CMV/A/C is used, set ventilator using Tidal Volume target and all other settings to remain the same. Only if severe ventilator dyssynchrony after adequate sedation and analgesia, paralyse with *~rocuronium 0.6mg/kg IVI~*
2. While Pressure-cycled SIMV is recommended, volume-cycled SIMV may be used if pressure-cycled SIMV is not available. If volume-cycled SIMV is used, set ventilator using Tidal Volume target and all other settings to remain the same.

Give mechanical ventilation general care:

- Start sedation and analgesia infusion (see table on page): give morphine and midazolam. If unstable, give ketamine instead.
- Wean FiO₂ unless hypotensive, acidotic or anaemic to target sats of 94%-98%.
- Unless hypotensive, raise head of bed 30°.
- Keep patient warm.
- Unless contraindicated (e.g. base of skull fracture) or will delay transport, insert nasogastric tube and place on free drainage.
- Reassess HR, BP, sats, ETCO₂, TV, Pplat, and auto-PEEP.

ARDS - Acute respiratory distress syndrome ARDS

I:E ratio - Inspiration : expiration ration

A/C - Assist-control ventilation

PEEP - Positive End-Expiratory Pressure

Pplat - Plateau pressure

CMV - Continuous mandatory ventilation

SIMV - Synchronized intermittent mandatory ventilation

IBW - Ideal body weight; height in cm - 100

TV - Tidal volume

ETCO₂ - End-tidal CO₂

- Continuously monitor for tube displacement/obstruction, cuff pressure, pneumothorax, equipment failure and secretions.
- Continue to treat underlying cause as per relevant page.
- Transport to closest emergency centre.
- Monitor ABCDE and pain and manage accordingly ↔PAIN.

Post-intubation analgesia and sedation dosing

If haemodynamically stable, use morphine and midazolam IV:

Drug and dose		Mixture and rate for long distance transportation	
Midazolam	• Loading dose: 0.05 to 0.1mg/kg • Maintenance: 0.06-0.24mg/kg/hr	45mg midazolam + sodium chloride 0.9% in 45mL total volume	Run at rate: 5-15mL/hr
Morphine	• Loading dose: 0.05-0.1mg/kg • Maintenance: 0.02-0.08mg/kg/hr	20mg morphine + sodium chloride 0.9% in 50mL total volume	Run at rate: 5-15mL/hr

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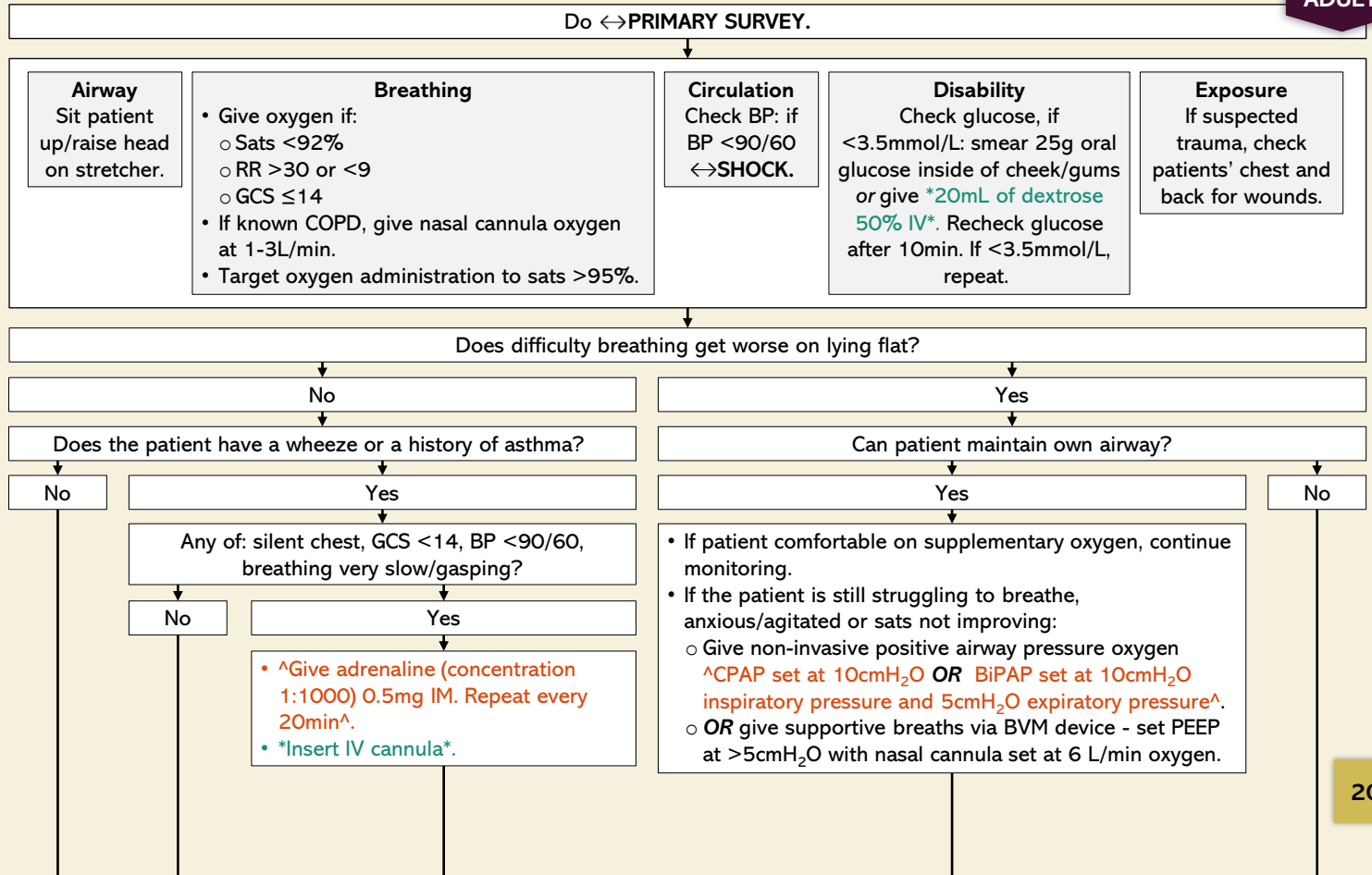
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Drug and dose		Mixture and rate for long distance transportation	
Ketamine 0.25-0.5mg/kg/hr		200mg ketamine + sodium chloride 0.9% in 50mL total volume	Run at rate: 5-10mL/hr
Midazolam 0.05mg/kg/hr		15mg midazolam + sodium chloride 0.9% in 50mL total volume	Run at rate: 15mL/hr

BREATHING DIFFICULTY



Use this tool for the adult (≥ 13 years) patient with difficulty in breathing, shortness of breath, wheeze or cough.



- *Give inhaled salbutamol 5mg and ipratropium bromide 0.5mg via inhaler or nebuliser. Repeat salbutamol continuously if still wheezing. Nebulise using in-line nebuliser attachment on BVM if manually ventilating patient.
- Insert IV cannula if not already done.
- *Give hydrocortisone 200mg IV*. or ^Give oral prednisone^
- If poor response to nebuliser, ^give magnesium sulphate 2g in 200mL sodium chloride 0.9% IV over 20min^.

- Any of: silent chest, GCS <14, BP <90/60, breathing very slow/gasping?

Yes

If not given already and if no response to above treatment, give ^adrenaline (concentration 1:1000) 0.5mg IM. Repeat every 20min^.

No

Limit patient distress:

- Reassure the patient.
- ^If CPAP/BiPAP, slowly increase cmH₂O until desired pressure^.
- Let the patient hold mask to their face before securing with straps.

Does the patient have swollen legs or a history of heart failure?

Yes

No

- *Insert IV cannula*.
- Is the BP >90/60?

Yes

No

- ^Give furosemide 40mg IV slowly. If no response after 30min, repeat to max 80mg.
- Give sublingual isosorbide dinitrate 5mg or GTN 0.4mg^.

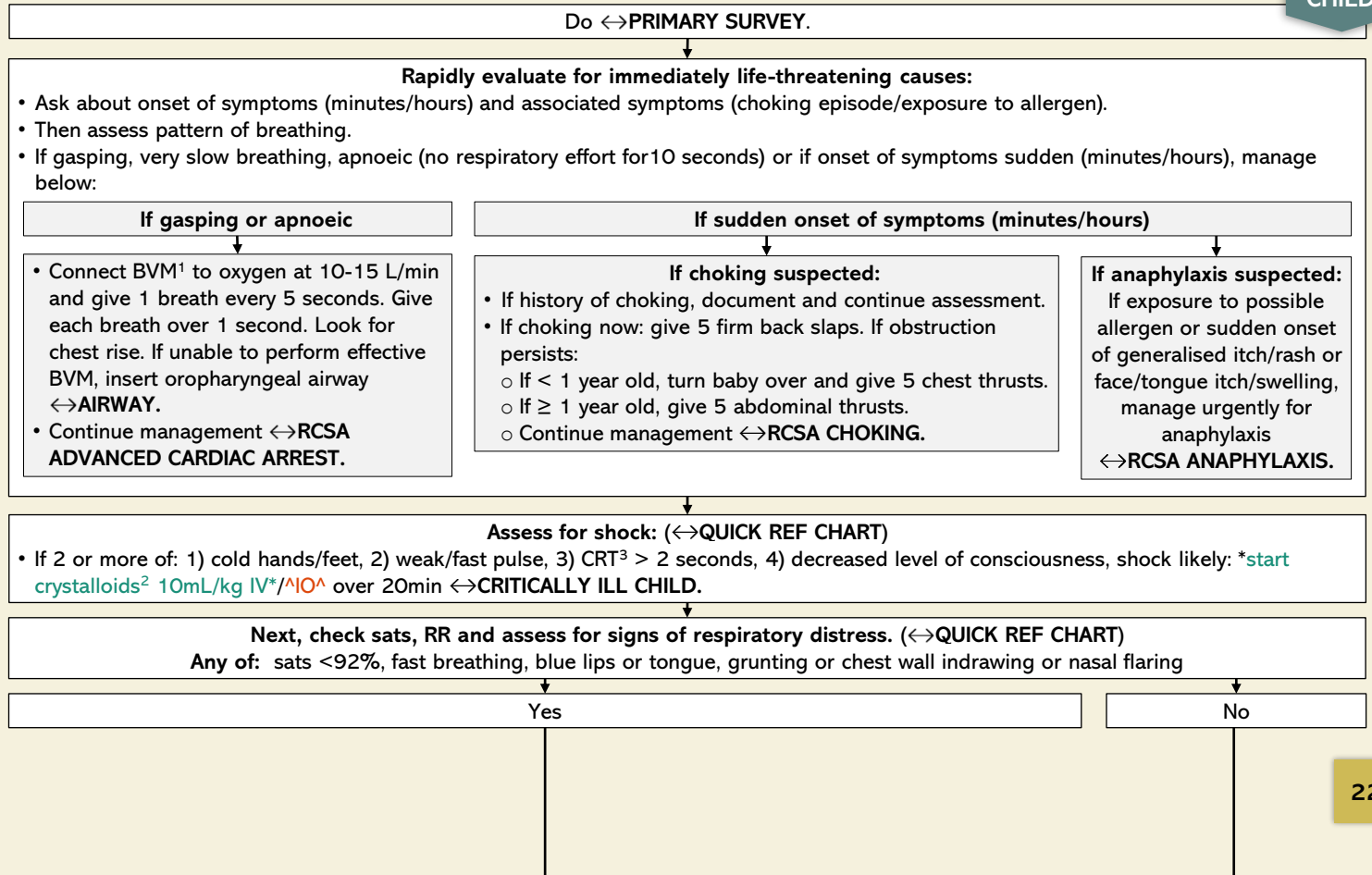
Transport to closest emergency centre

If patient cannot maintain own airway, sats <92%, breathing very slow/gasping ↔ **AIRWAY**.

- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP <90/60, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation.
- Manage underlying conditions.
- Manage pain ↔ **PAIN**. If airway unsecured, avoid high analgesic doses as these have respiratory depressant effects.

BREATHING DIFFICULTY

Use this tool to manage an infant or child, 1 month to 13 years old, presenting with breathing difficulty.



- Give oxygen at 1-2L/min via nasal cannula. If sats <92%, give oxygen at 4-6L/min via facemask.
- Check glucose: if <3.5mmol/L: ***give dextrose 10% 3mL/kg IV*/¹IO¹** as a bolus and recheck glucose after 10min. If no IV access: smear 25g glucose gel inside cheeks/gums or ***give glucagon 0.5mg IM***.
- If stridor or wheeze, manage below:

If stridor:

- **²Give nebulised adrenaline 0.5mL/kg of 1:1000 adrenaline. (max 5mL).**²
- Assess response: if needed, repeat every 10-15min.
- Connect to 3-lead ECG if available
- Monitor for development of tachycardias and arrhythmias – if occur contact receiving facility for advice.

If wheeze:

- Give 10 puffs of salbutamol via metered dose inhaler (MDI) and spacer. If < 3 years old, use mask attached to spacer.
- If no MDI available, or continuous facemask oxygen needed: give instead nebulised B2-agonist (salbutamol or fenoterol):
 - If < 3 years old: add 0.5mL B2-agonist to 4mL normal saline.
 - If ≥ 3 years old: add 1mL B2-agonist to 4mL normal saline.

- If poor response or sats < 92%, give continuous nebs.
 - Contact receiving facility for additional advice.
 - If known history of asthma:
 - Add nebulised ipratropium bromide
 - **²Give oral prednisone² OR ²IV hydrocortisone².**

Expedite transfer to nearest appropriate facility

- If severe distress, pre-notify the facility of your arrival.
- Check BP, HR, RR, sats, glucose and AVPU regularly.
- If worsening difficulty breathing or very slow breathing: if trained personnel and resources available, start non-invasive ventilation (mask CPAP or high flow nasal oxygen) ↔ **AIRWAY**.

Transport to nearest emergency centre or most appropriate facility

- Monitor and record BP, HR, RR, sats, and GCS every 10min. Reassess and manage airway, breathing, circulation according to age appropriate vital signs in child ↔ **QUICK REF CHART**.
- Obtain history and document: history of possible toxin ingestion/inhalation, history of cardiac disease.

¹ Bag mask ventilator device – ensure correctly sized equipment.

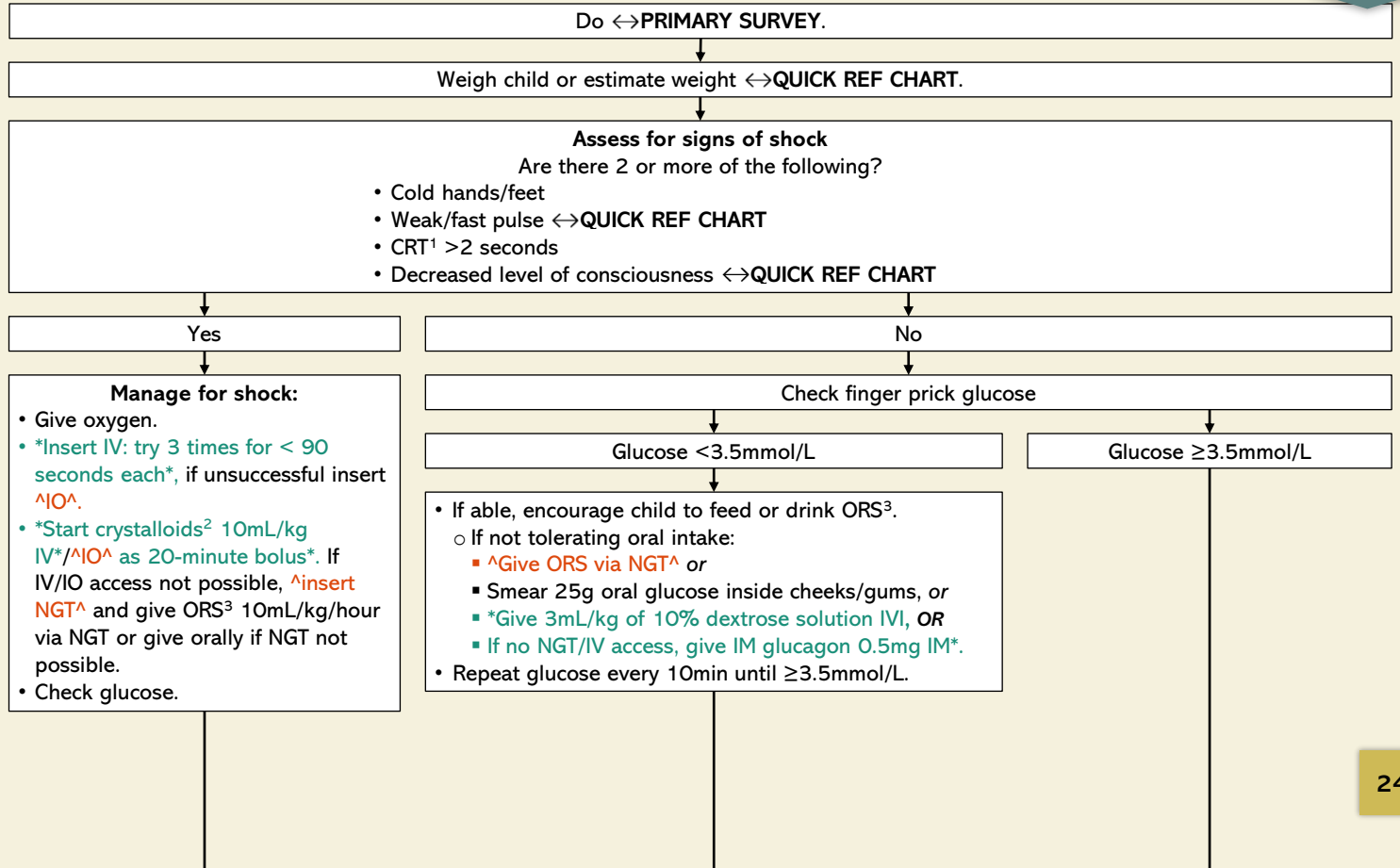
² Balanced crystalloids e.g. Ringers Lactate, plasmalyte.

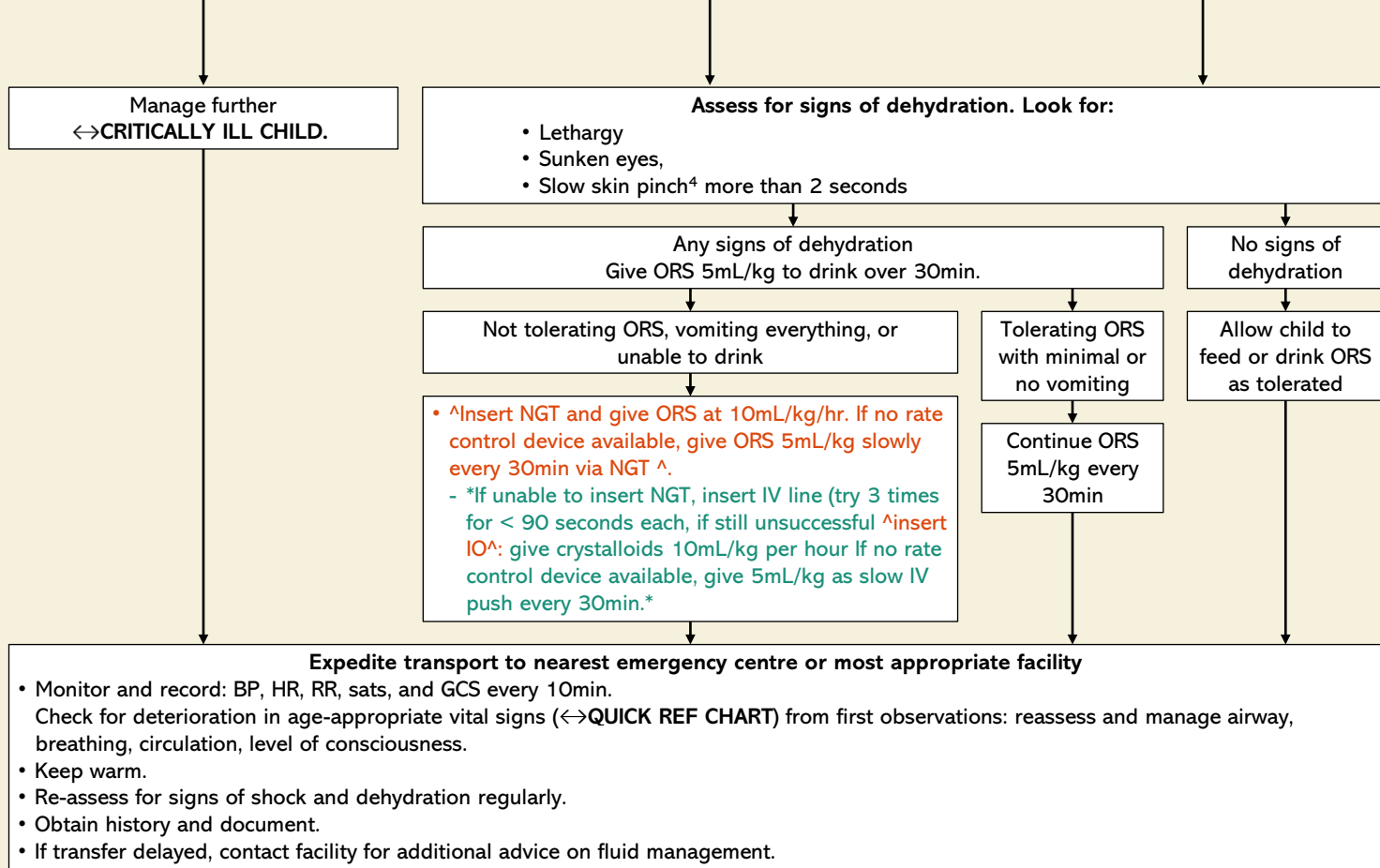
³ Capillary refill time (CRT): hold hand/foot higher than level of the heart, press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return.

CHILD WITH DIARRHOEA



- Use this tool to manage the infant/child (1 month old – 13 years old) with diarrhoea, with or without vomiting.
- If poisoning suspected ↔ **POISONING**.





¹Capillary refill time (CRT): hold hand/foot higher than level of the heart. Press soft pad of finger/toe until it turns pale, then release pressure and note time taken for colour to return.

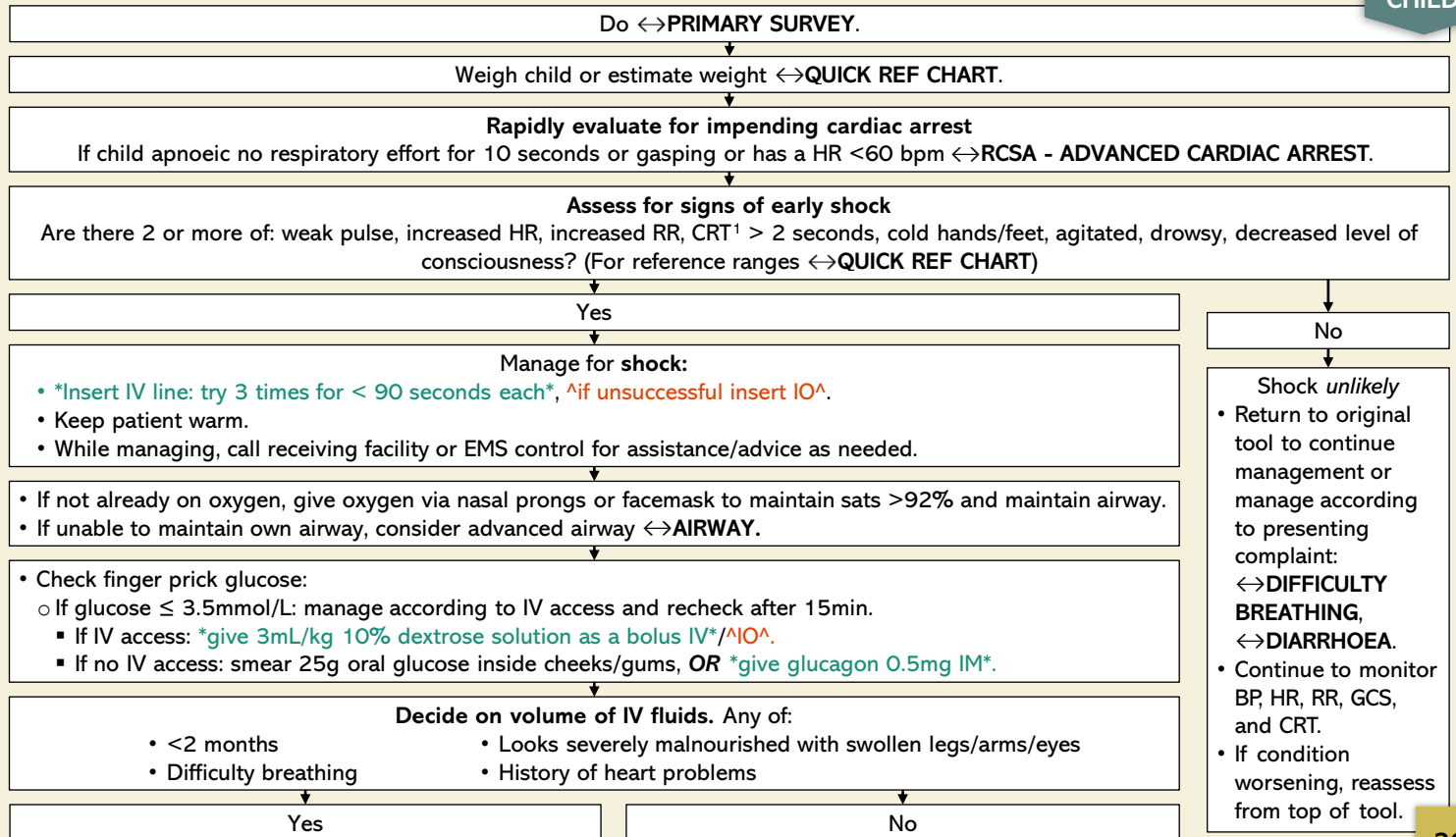
²Ideally balanced crystalloids e.g. Ringers Lactate, Plasmalyte.

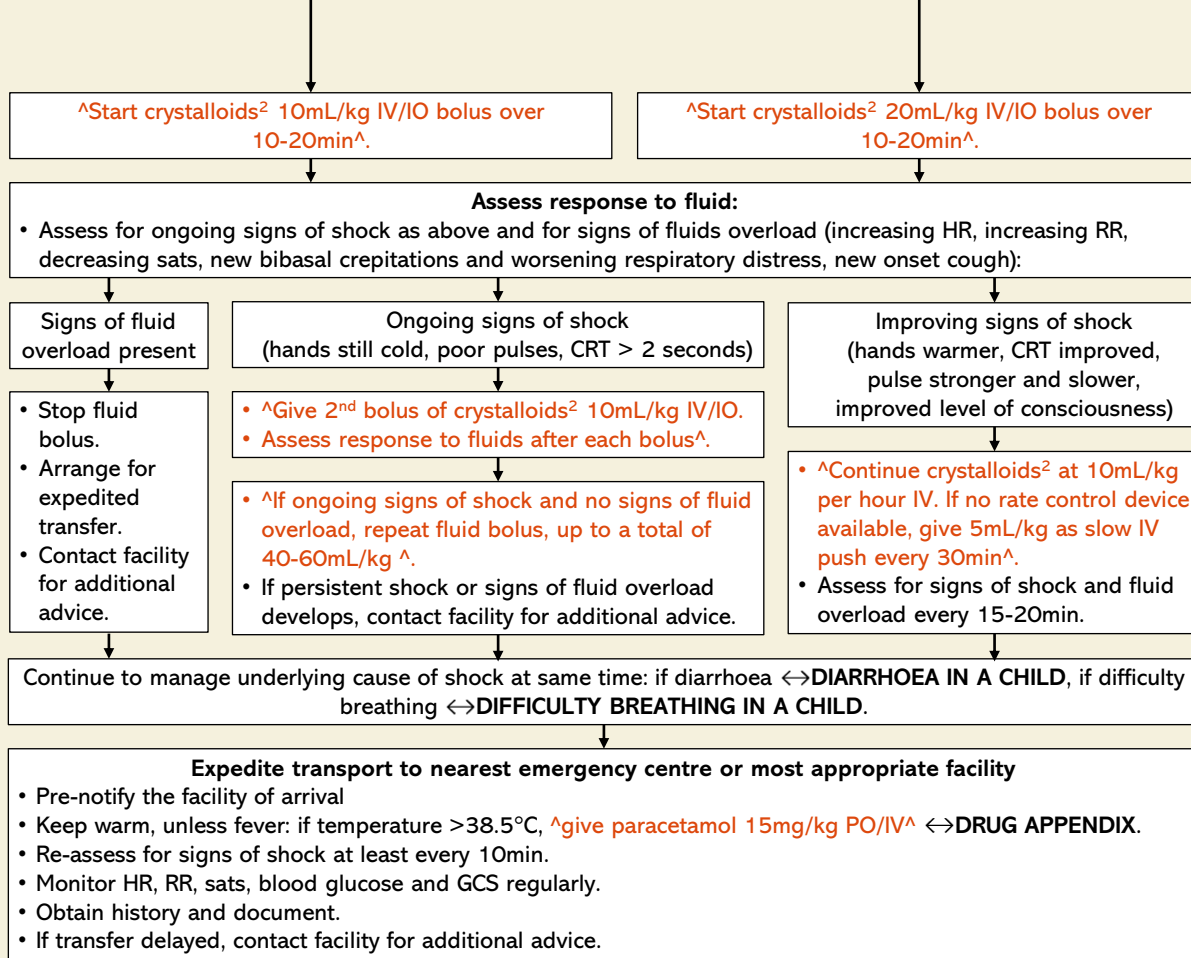
³Oral Rehydration Solution - dissolve 8 flat teaspoons of sugar and ½ teaspoon salt in 1L sterile water.

⁴Pinch skin on abdomen between 2 fingers, release - skin usually snaps rapidly to its normal position - a slow skin pinch takes longer to return to normal.

APPROACH TO THE CRITICALLY ILL CHILD

Use this tool to manage the infant/child -1 month old – 13 years old, and in any critically ill or injured child.





¹Capillary refill time (CRT): hold hand/foot higher than level of the heart. Press soft pad of finger/toe until it turns pale (press for around 5 seconds), then release pressure and note time taken for colour to return. Alternatively, check CRT over sternum.

²Ideally balanced crystalloids e.g. Ringers Lactate, Plasmalyte (NOT dextrose containing fluids).

³ IV fluid administration: note scope of practice: *child >1year: AEA, ECT, ECA, ANT, ECP*; ^child <1year ECT, ANT, ECP^

BADLY INJURED ADULT PATIENT



ADULT

Use this algorithm to conduct a rapid trauma survey for the adult patient (≥ 13 years old) who presents with any injury (or injuries/polytrauma) that have the potential to be life-threatening and that may require immediate intervention.

Do $\leftrightarrow$ **PRIMARY SURVEY.**

Then perform a rapid trauma survey.

Limit on-scene time: perform only life-saving and stabilising interventions.

If available, call for additional resources early.

Airway

- If airway obstructed (snoring, gurgling, noisy breathing), open using jaw thrust, keeping neck stable. Avoid head-tilt and chin-lift.
- If foreign body is visible in the mouth: remove **using Magill's forceps** if needed.

If unable to maintain airway, connect BVM to oxygen and slowly deliver¹ breath every 4-6 seconds.

Breathing

- Assess work of breathing and estimate RR.
 - If difficulty breathing or sats $< 92\%$, give face mask oxygen at 15L/min.
 - If very slow RR/gasping/blue lips/tongue, connect BVM to oxygen and slowly deliver each breath with the patient (1 breath every 4-6 seconds).

- If breathing problems persist despite using BVM and $\leftrightarrow$ **AIRWAY.**

Circulation

- Assess pulse: if no pulse, start CPR $\leftrightarrow$ **RCSA TRAUMA ARREST.**
- Attach monitors: sats, HR and BP.

- Stop visible bleeding - apply pressure and elevate limb. Perform wound packing with haemostatic gauze. If bleeding still severe, apply tourniquet above injury.

Disability & Exposure

- If head/spine injury, use log-roll to turn and $\leftrightarrow$ **SPINAL INJ.**
- If seizures $\leftrightarrow$ **SEIZURES.**
- Assess level of consciousness using AVPU or GCS scale.
- Check glucose: if $< 3.5\text{mmol/L}$:
 - Smear 25g oral glucose on inside of cheek/gums or,
 - **Give 20mL of dextrose 50% IV**
 - If no IV access, give glucagon 1mg IM.
 - If known alcohol user, give thiamine 100mg IV (or IM!) before glucose.*
 - Recheck glucose after 10min. If $< 3.5\text{mmol/L}$, repeat.
- Keep patient warm.

• If unable to perform effective BVM ventilation, insert OPA and ↔**AIRWAY**.

• If open chest wound, apply a three-way dressing and observe for tension pneumothorax.
• Tension pneumothorax likely if open chest wound and 1 or more of the following are present: sudden breathlessness, more resonant/ decreased breath sounds/ pain on one side, deviated trachea, SBP <90.
○ *Insert large bore cannula in the 5th intercostal space rib in anterior-axillary line*. Observe for signs of recurrence.

• If SBP <90 or HR >100 or visible active bleeding or decreased GCS:
○ *Insert x1 large bore IV cannula. If no delay caused, insert second IV cannula.
○ Give crystalloid fluids 100-250mL IV rapidly.*
○ Titrate volume resuscitation to maintain a palpable carotid or femoral pulse. Do not use permissive hypotension in pregnancy, TBI, the aged or the very young.
○ If head injury, adjust fluid resuscitation to maintain cerebral perfusion, MAP >80mmHg.
• If pregnant patient >20 weeks, perform left lateral tilt 15-30° or manually displace uterus.
• If active bleeding, give [^]tranexamic acid[^] 2g IV over 10min.

• Consider cause: if catastrophic haemorrhage with shock ↔**SHOCK**, if head injury ↔**HEAD INJURY**, if spinal cord injury suspected ↔**SPINAL INJ**, if burns ↔**BURNS**.
• Reassess ABCDE frequently to identify deterioration/change early.

• If the patient continues to deteriorate despite life-saving interventions, uncontrollable bleeding, or intractable pain, expedite transfer to closest emergency centre or most appropriate facility (and pre-notify facility of arrival).
• If expedited transfer not required, do a head-to-toe secondary examination ↔**TRAUMA SURVEY**.

Transport to closest emergency centre or most appropriate facility:

• Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP <90/60, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation.
• Assess and manage pain ↔**PAIN**.
• Manage underlying conditions.
• Obtain history and document: ask about allergies, medications, past medical history, last meal, events prior to injury, mechanism of injury.
• [^]Perform eFAST¹ to obtain cardiac, thoracic, abdominal and pelvic views: look for evidence of haemopericardium (with or without tamponade), pneumothorax, haemothorax, or free fluid in abdomen or pelvis[^]. (keep transport times in mind)

¹eFAST (extended focused assessment with sonography for trauma).

TRAUMA SECONDARY SURVEY



Use this algorithm to manage the patient who presents with any injury (or injuries) after performing the rapid trauma survey.

Do ↔PRIMARY SURVEY.

Then perform a rapid trauma survey ↔BADLY INJ.

Next, do a head-to-toe examination

	Inspect	Palpate/auscultate	Note
General	Look for medical alert devices.		
Head and face	• Look for deformity, lacerations, bruising, asymmetry. • Check pupils. • Check eyes, ears and nose for bruising around eyes (raccoon eyes¹) or ears; or blood or clear fluid leaking from nose or ear. • Check mouth for airway/tongue swelling and loose teeth. • Assess and record level of consciousness using GCS.	Run fingers over scalp without applying excessive pressure: feel for crepitus ² , bony tenderness, subcutaneous emphysema ³ .	If head injury or raccoon eyes, cerebrospinal fluid (CSF) leaking from nose or ears, pupils unequal or respond poorly to light ↔ HEAD INJURY .
Neck	Check for deformity, lacerations, bruising.	• *Feel c-spine for mid-line tenderness*. • Feel for carotid pulse. • Feel for tracheal position. • Feel for subcutaneous emphysema.	• If c-spine pain, ensure spinal precautions ↔SPINAL INJ. • If deviated trachea or subcutaneous emphysema, check for pneumothorax.
Chest	Check expansion, paradoxical movement, accessory muscle use, lacerations, deformity.	• Feel for tenderness, subcutaneous emphysema, crepitus, apex beat. • Listen to heart sounds, air entry and breath sounds.	• If rib fractures suspected, check for pneumothorax. • If flail chest⁴, given high flow oxygen and pain relief ↔PAIN. • If wheeze ↔BREATHING DIFFICULTY ADULT.

	Inspect	Palpate/auscultate	Note
Abdomen	Check for lacerations, bruising, distension.	• Feel for tenderness, acute abdomen (guarding rigidity, rebound tenderness⁵). • Listen for bowel sounds.	If lower abdominal pain with blunt trauma, apply a pelvic binder.
Pelvis	• Check for deformity, lacerations, bruising. • Check for urinary/faecal incontinence. • Check for priapism (spinal injury).	Feel for bony tenderness.	If tender or dangerous mechanism of injury, apply a pelvic binder and spinal motion restriction.
Upper and lower limbs	Check for deformity, lacerations, bruising, shortening or rotation.	• Check for bony tenderness, weakness/numbness. • If wound, check for poor perfusion (cold, pale, numb, no pulse) below injury.	If fracture suspected: • if bleeding proximal to deformity, control. • splint limb. • if poor perfusion, document.
Back	• Log roll patient to check for deformity, lacerations, bruising. • Screen for obvious focal neurology.		Restrict spinal motion.

Transport to closest emergency centre.

- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP <90/60, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation.
- Manage underlying conditions.
- Manage pain ↔ **PAIN**. If airway unsecured, avoid high analgesic doses as these have respiratory depressant effects.

¹Raccoon eyes refers to bruising around orbits suggestive of base of skull fracture.

²Crepitus refers to the crackling, popping, or grating sound or feeling that can be heard or sensed when certain movements are made.

³Subcutaneous emphysema refers to the presence of air in the layer of tissue beneath the skin.

⁴Flail chest refers to chest injury in which two or more ribs become detached from the rest of the ribcage.

⁵Rebound tenderness refers to clinical sign where pain is increased when pressure on the tender area is released.

DROWNING



Use this tool to assess and manage the patient who is in, or recently recovered from a water submersion incident.

Check for and address hazards to rescuer and patient.

- Ensure wearing correct PPE.
- Assess risk of hypothermia due to water temperature.
- Assess mechanism of injury for potential spinal injury.
- Assess risk of contaminants or polluted water.
- Assess risk to rescuer from strong currents or swift water.

Has patient been removed from the water?

No

- Remove patient from water without risk to rescuer or patient:
 - Reach: if in arms reach, reach for patient.
 - Throw: if not in arms reach, throw floatation device to patient.
 - Row: use watercraft to reach patient. Wear a lifejacket.
 - Go: only if trained in water rescue and rescuer is not at risk.
- Remove wet clothes. Dry and cover with blankets. Protect from wind.

Yes

Is Airway open/patent and self-maintained?

No

- Check airway for and remove foreign bodies.
- Turn patient lateral. Log roll lateral if injured.
- If c-spine injury suspected, maintain c-spine control and use jaw thrust to open airway.
- If no c-spine injury suspected, use head-tilt chin-lift to open airway or turn patient lateral.

Yes

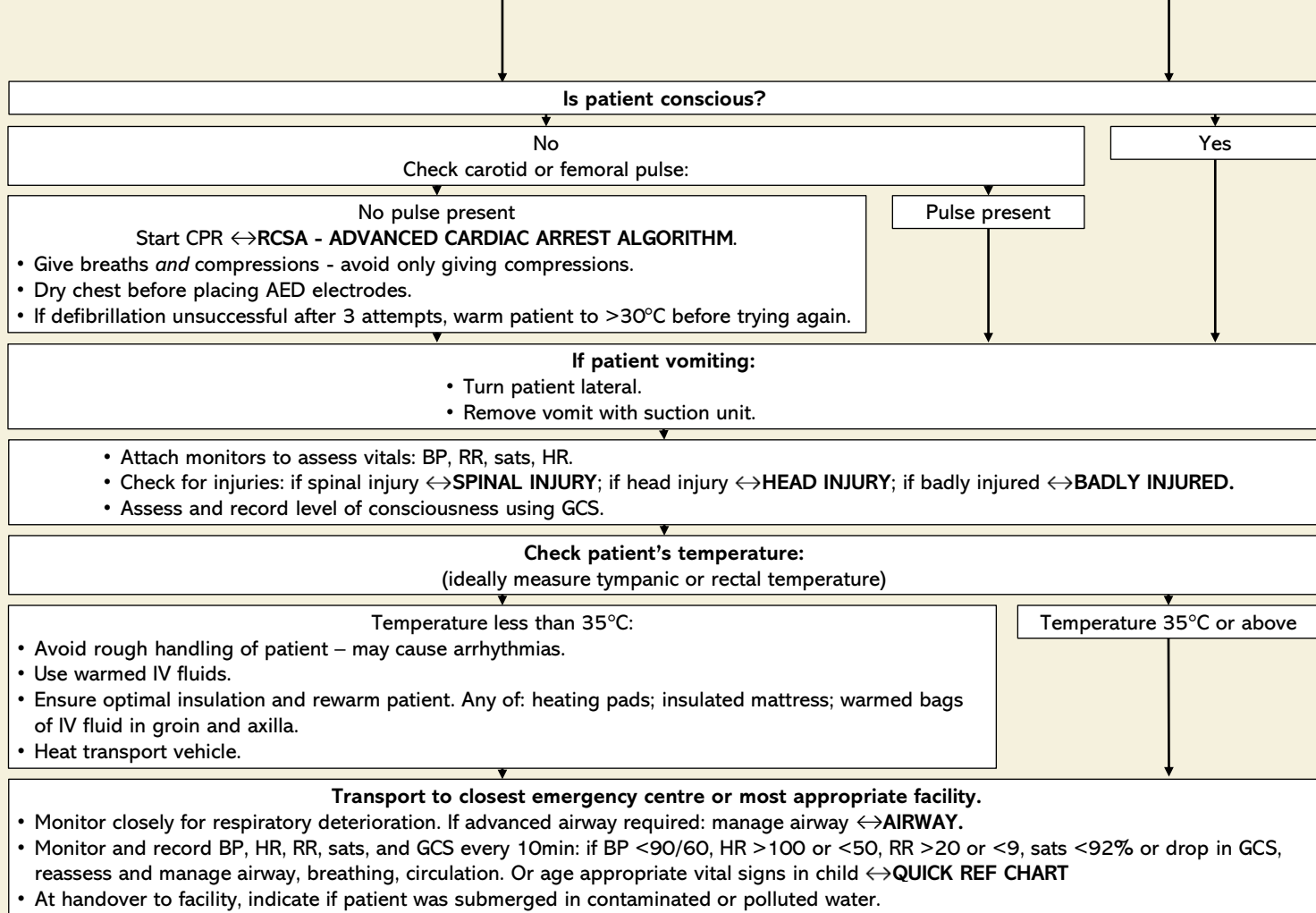
Are there signs of Breathing?

No

Respirations absent, very slow or patient gasping:

- Give 5 rescue breaths with oxygen.
- Then assist breathing using BVM to give 1 breath every 5 seconds.

Yes



BURNS



Use this tool for the patient who has been injured in a fire.

Stop burn process: smother flames with blanket and move patient away from burn source. Remove hot/burnt and tight clothing.

Do ↔ **PRIMARY SURVEY**.

Airway: check for signs of inhalational injury

- Burns to face or nasal hairs, eyebrows or eyelashes.
- Black soot around nose or mouth.
- History of being trapped/unconscious/removed from burning building.
- Coughing up black sputum or has a hoarse voice (stridor).
- Unable to maintain own airway or difficulty breathing.

Breathing: attach sats monitor, check GCS and RR, obtain basic history. If any of the following, give 100% non-rebreather mask oxygen:

- GCS ≤ 14
- Sats $< 92\%$
- RR > 30
- RR < 9
- Suspected carbon monoxide or cyanide poisoning (headache, dizziness, vomiting, chest pain, confusion).
- Involved in fire in an enclosed space.

Any of: inhalation injury, or patient unable maintain own airway, or sats $< 92\%$, or breathing very slow/gasping despite 100% oxygen?

Yes ↔ **AIRWAY** + ^Nebulised Adrenaline 1-5ml of 1:1000^ if upper airway involvement is suspected

No

- Start to cool burn with cool running tap water or wet towel/s for a least 20min (1 hour if chemical burns).
 - If no clean water available, use hydrogel products. Stop if patient starts to shiver.
- Assess depth of burn wound (see reference guide below).
- Assess percentage of total body surface area (TBSA) burnt using chart.
- Cover burn with loose and light dressing, like plastic wrap, clean dry sheet or blanket.

Circulation: assess for shock according to age of patient:

ADULT patient:
SBP < 90 (or MAP < 65)?

Yes

No

CHILD patient (< 13 years):

Check pulse, CRT and BP (↔ **QUICK REF CHART**). Any of:

- Altered mental status
- Fast respiratory rate for age
- Fast pulse for age
- Low SBP according to age
- Weak peripheral pulses
- Delayed CRT (> 2 sec)

Yes

No

- Search for other causes of hypotension ↔ **BADLY INJURED**.
- Give 250-500mL crystalloid fluid IV bolus over 5-10min.
- Reassess BP: if SBP <90 (or MAP <65), repeat.
- Stop fluids if breathless or new crackles on auscultation.
- Once IV fluids are running proceed to next step.

- Signs of shock
- Give fluids ↔ **CRITICALLY ILL CHILD**.
 - Once fluid resuscitation complete:

No signs of shock

Continue to calculate fluid needs according to TBSA of burn:

Burns ≤10% TBSA for children or ≤20% TBSA for adults

Burns >10% TBSA for children, or >20% TBSA for adults

- If not already done, ***attempt IV access*** ^{^(if unable to obtain IV access, consider IO access)^}.
- Calculate IV fluid needs using Parklands Formula:
 - [Total % burn X weight (kg) X 4mL]
 - Give half of this amount over first 8 hours calculating time from injury, and other half over the following 16 hours.
- If patient was fluid resuscitated first, subtract the volume of fluid given during resuscitation from the Parklands Formula total. Continue giving remainder.

- Once IV fluids are running, assess and manage pain ↔ **PAIN**.
- Check for other injuries: if badly injured ↔ **BADLY INJURED**.
- Maintain temperature above 36°C, use dry sheets and blankets.

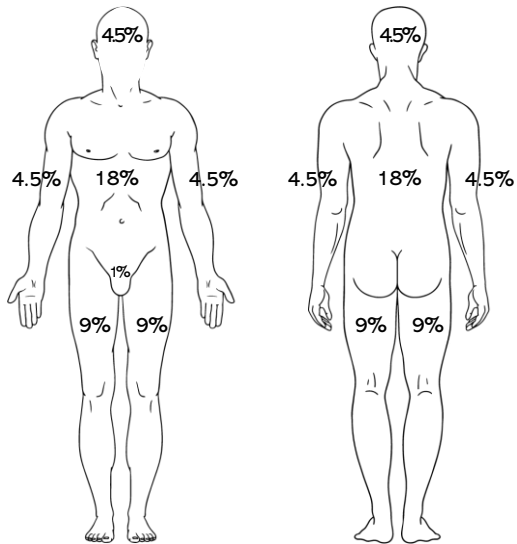
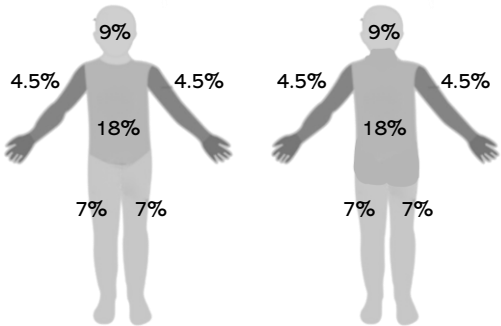
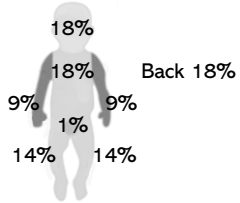
Determine most appropriate receiving facility. If any of the following, consult with senior medical advisor:

- Partial thickness burns >10% TBSA
- Full thickness burn >3% TBSA
- Any burn involving head, face, hands, feet or perineum
- Inhalation injury
- Circumferential burns
- Electrical burn
- Burns in neonate or child
- Serious pre-existing or concomitant injuries

Transport to most appropriate facility

- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP <92, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation. Or age-appropriate vital signs in child ↔ **QUICK REF CHART**.

Assess depth of burn:	
Depth of burn	Signs and symptoms of burnt area
Superficial or epidermal	Dry, minor blisters, redness, painful.
Superficial partial thickness or superficial dermal	Moist, listers, painful.
Deep partial thickness or deep dermal	Moist, white or yellow slough, red mottle, less painful.
Full thickness (complete loss of dermis)	Dry, charred whitish, brown or black, painless .

Assess body surface area of burn:		
Adult surface area	Child surface area	Infant
		

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SUSPECTED SPINAL INJURY FOLLOWING TRAUMA



Use this tool where there is concern about spinal injury following trauma.

- Evaluate scene safety: consider fire, contamination, traffic, open water, weapons or active shooting, unstable structures.
 - If not safe to approach, evaluate for hazards and activate appropriate resources (like fire and rescue services, traffic police, mountain rescue).
- If possible, assign someone to crowd control to reduce public view of incident.
- In penetrating or time critical injuries (e.g. bleeding), the priority is rapid transport. Do not delay scene time to apply spinal motion restriction.

Do ↔ **PRIMARY SURVEY.**

Is patient in a vehicle?

Yes

- If unconscious: check pulse and breathing: if pulseless/not breathing, rapidly extricate from vehicle if safe for provider.
- If conscious: assess whether patient trapped:

Patient trapped

- Activate appropriate resources (e.g. fire and rescue services).
- If conscious, ask patient's name and reassure/calm patient.
- Once freed, see if the patient can self-extricate.
- If unable to self-extricate, use manual in-line stabilisation of c-spine and extricate with a trauma long-board.

Patient not trapped

- Ask patient's name and encourage self-extrication unless confused or unable to stand.
- If unable to self-extricate, use manual in-line stabilisation of c-spine and extricate with best available resources (spine board, scoop, vacuum mattress, etc.).
- If infant in baby seat/capsule, leave baby in seat and remove seat.

No

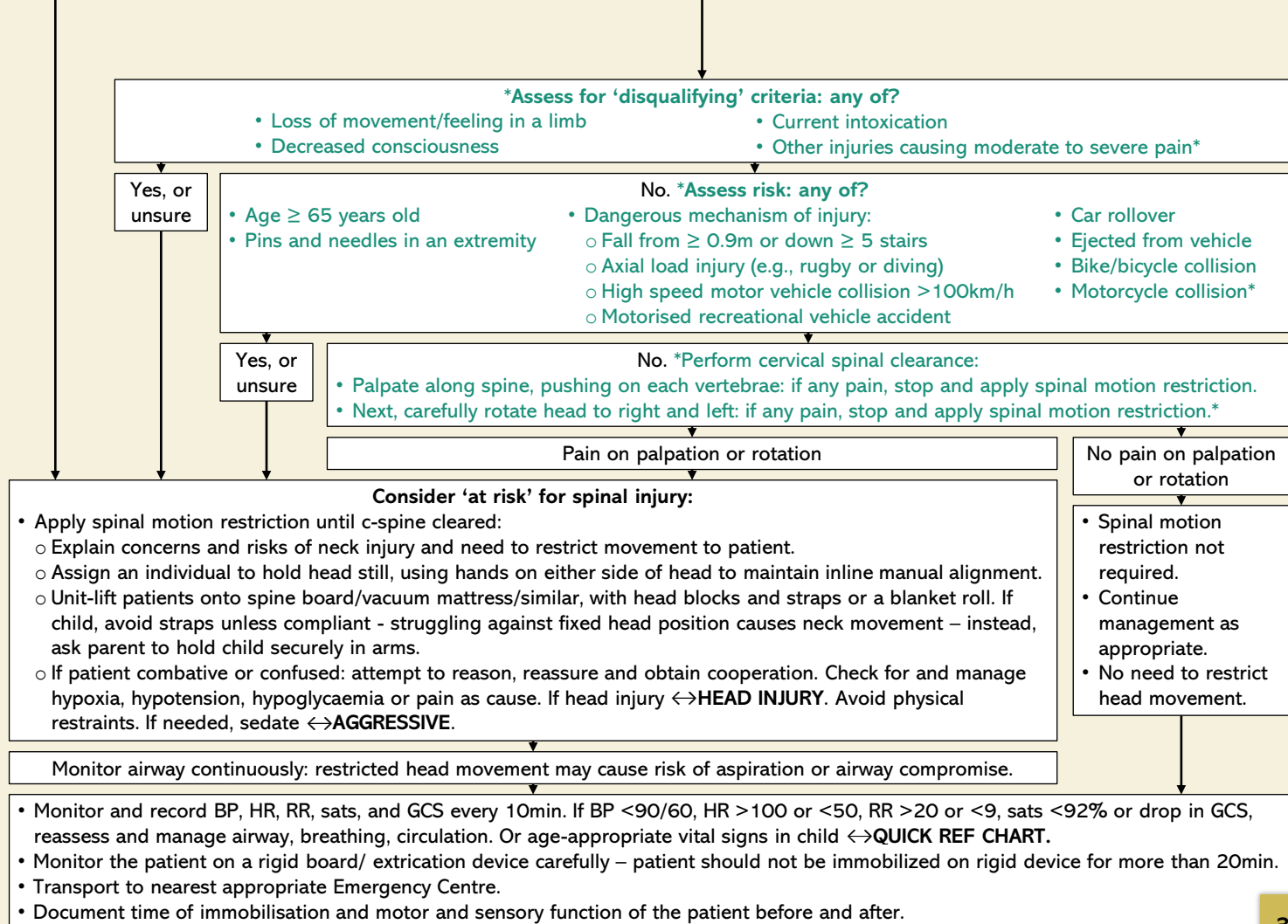
- If safe, manage on scene where found.
- If not safe, move patient to safety using best possible spinal immobilisation method to minimise spinal movement.

During extrication, explain all actions and where appropriate, update patient on safety of others involved in incident.

- Perform rapid trauma survey ↔ **BADLY INJURED PATIENT.**
- Manage further according to level of training: are you trained to perform cervical spine clearance?

No

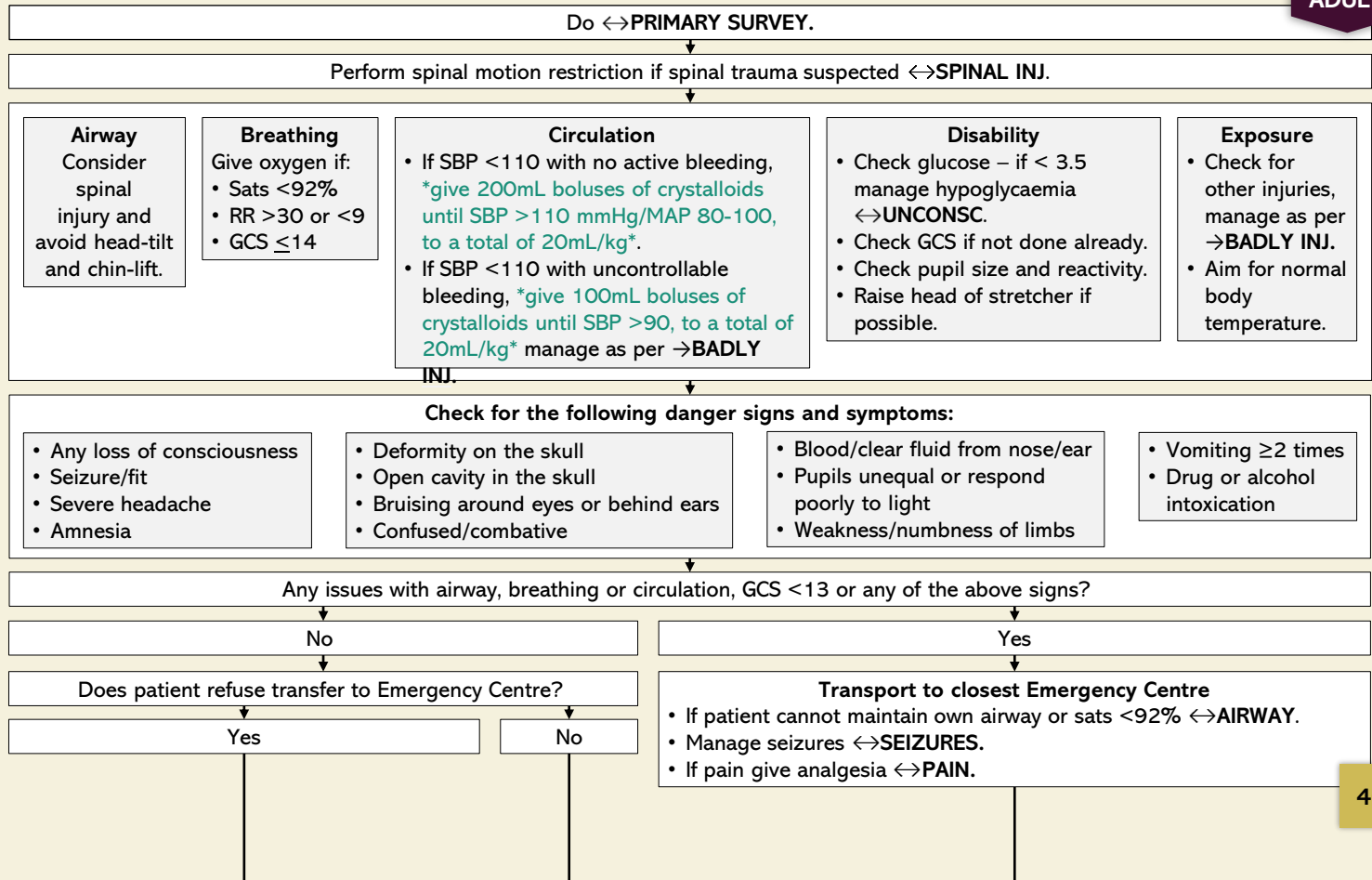
Yes



HEAD INJURY



Use this tool for the adult patient (≥ 13 years) who has sustained injury following physical force to the head.



↓

If mild headache, dizziness, forgetfulness, lack of focus, **advise patient:**

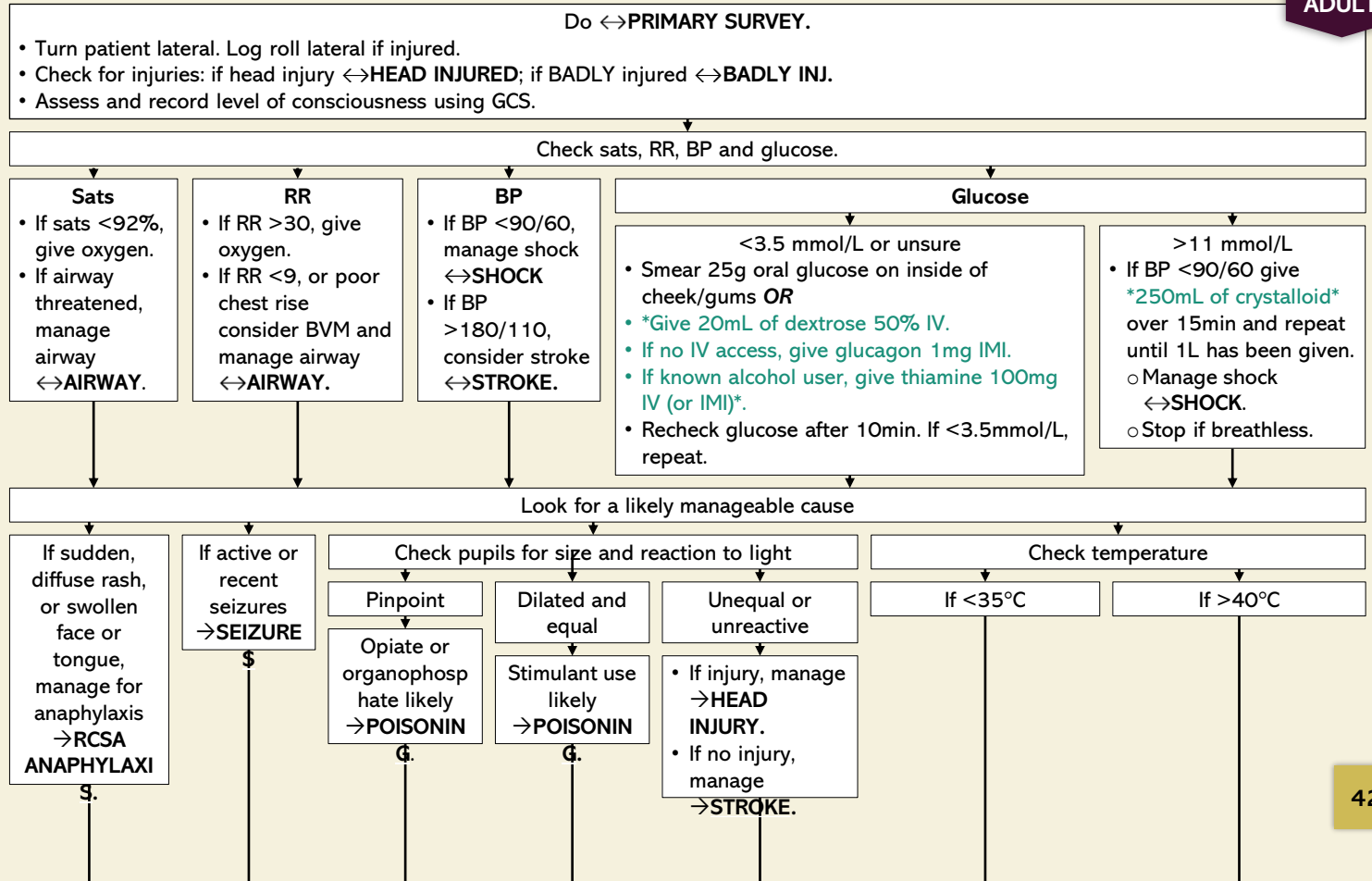
- To rest for 2 days.
- If no symptoms >3 days, gradually increase exertion.
- If not recovered by 6 weeks, to seek care.
- Call EMS or go to hospital immediately if any of above danger signs and symptoms occur.

- ↓
- Manage body temperature to keep within normal range.
 - If ventilating with BVM or mechanical ventilator maintain ETCO_2 35-45mmHg.
 - Monitor BP, HR, RR, sats, and GCS every 10min.
 - If BP <90/60, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation, level of consciousness.

DECREASED CONSCIOUSNESS IN ADULTS



Use this tool for any patient not fully orientated and conscious (patient ≥ 13 years old).



- Avoid rough movements.
- Remove cold/wet clothes; dry and cover with blankets.
- If possible warm IV fluids.
- If $<32^{\circ}\text{C}$ give warm fluids (40 to 42°C) if possible.
- Transport in heated vehicle.

- Remove clothes.
- Place ice packs, towels in axilla, neck and groin. Or cover body with wet towels.
- Transport with aircon if possible.
- Stop cooling if temperature $<39^{\circ}\text{C}$.

Transport to closest emergency centre or most appropriate facility.

- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP $<90/60$, HR >100 or <50 , RR >20 or <9 , sats $<92\%$ or drop in GCS, reassess and manage airway, breathing, circulation.
- Manage underlying conditions.
- Manage pain $\leftrightarrow$ **PAIN**.

SEIZURES/FITS



**ADULT &
CHILD**

Use this tool if managing the patient who is fitting/having a seizure or has just had a fit/seizure (including focal seizures).

Do ↔ **PRIMARY SURVEY.**

- Check glucose: if $<3.5\text{mmol/L}$: smear 25g oral glucose on inside of cheek/gums or, give ***20mL of dextrose 50% IV or if child ($<13\text{yrs}$), give 3mL/kg dextrose 10% IV. If no IV access, give glucagon 1mg IM (child 0.5mg)*. Recheck glucose after 10min. If $<3.5\text{mmol/L}$, repeat.**
- Check if patient could be pregnant >20 weeks – if pregnant and not a known epileptic, treat as → **ECLAMPSIA.**

Is patient still fitting?

Yes

No

- Protect patient from injury: provide safe space – mattress/ blanket, lateral side, do not restrain.
- Clear airway, suction if secretions and administer high-flow face mask oxygen.
- Is IV access rapidly and safely available?

No IV access

IV access available

ADULT

- *Give midazolam 10mg by IMI, buccal, or intranasal.
- If not available give:
 - lorazepam 4mg IMI **OR**
 - diazepam 10mg PR*

CHILD ($<13\text{yrs}$)

- *Give midazolam 0.5mg/kg buccal **OR** 0.15mg/kg IMI **OR** 0.2mg/kg intranasal
- If not available:
 - Give diazepam 0.5mg/kg PR **OR**
 - Lorazepam 0.1mg/kg buccal*

NEONATE (<28 days)

See drug dosing charts.

ADULT

- Give one of the following slowly over 2min:
 - *midazolam 10mg IV **OR**
 - lorazepam 4mg IV
 - If not available give diazepam 10mg IV*.

CHILD (<13 yrs)

- Give one of the following slowly over 2min:
 - *midazolam 0.2mg/kg IV **OR**
 - diazepam 0.25mg/kg IV **OR**
 - lorazepam 0.1mg/kg IV*

NEONATE (<28 days)

See drug dosing charts.

Is patient still fitting 5min after first dose?

Yes

No

- Repeat dose of same medication (same route or IV if success at access).
- Monitor airway and breathing: if RR <9 or sats <92%, support ventilation with bag-valve mask (BVM).
- Is patient still fitting 10min after second medication?

Yes

No

- Call for medical advice and expedite transfer.
- Monitor and manage airway and breathing to maintain sats >92%.

Complete secondary survey

- Look for and manage injury ↔**HEAD INJ**; ↔**BADLY INJ**.
- Turn patient into lateral position to protect airway.
- If patient cannot maintain own airway, sats <92%, breathing very slow/gasping ↔**AIRWAY**.
- Check BP, in adult if <90/60 ↔**SHOCK**, child look for signs of shock →**CRITICALLY ILL CHILD**.

Give further care once the fit has stopped

- Reassure family and explain that patient needs hospital assessment and care as fits may recur and to find cause of fit.
- Consider establishing IV access if hospital >10min away and: repeat fit likely, known with epilepsy, previous repeated fits or untreated cause of fits.
- If unable to establish IV access, consider IO access in patients with prolonged transport time.

Transport to closest emergency centre.

- Monitor and record: BP, HR, RR, sats, and GCS every 10min
 - Adult: Check BP <90/60, HR >100 or <50, RR >20 or <9, sats <94% or drop in GCS, reassess and manage airway, breathing, circulation, level of consciousness.
 - Child: Check for deterioration in age-appropriate vital signs from first observations: reassess and manage airway, breathing, circulation, level of consciousness (see **QUICK REF CHART**).
- Document
 - Ask if any previous fits, epilepsy, medications, any substance misuse or possible poisoning?
 - Note times of fits and responsiveness, including fit onset, progression, duration.
 - Note all medications given, routes, doses and times (as well as response to medications).

AGGRESSION OR AGITATION



ADULT &
CHILD

Use this tool for the patient who is a danger to themselves, others and surroundings and not obeying commands.

Ensure the safety of all EMS personnel and the patient.

Be vigilant for unpredictable behaviour. Contact SAPS for help if the potential risk to the patient, family or staff is significant. Obtain a brief history from the family/friends, if possible, before you approach the patient. If restraint is considered, SAPS to be contacted

First try to convince the patient to calm down and co-operate.

Try to:

- Approach the patient from a safe distance
- Talk in a quiet, calm area if possible
- Listen to the patient's concerns
- Be calm and reassuring
- Show kindness, respect and empathy
- Set clear boundaries regarding the interaction

Avoid:

- Turning your back or allow yourself to be cornered
- Making direct eye contact
- Touching the patient unnecessarily
- Arguing or confronting any delusions
- Attempting to reason with the patient
- Engaging in prolonged conversation

Is the patient still aggressive/severely agitated?

Yes

No

This patient should be sedated. Is the patient willing to accept voluntary sedation for the transfer?

No

Yes

- Restrain physically and prepare to sedate the patient.
- Explain to the patient what is about to happen.
- Remove weapon/mechanism for self-harm.
- Restrain with help of 5 people: one restraining each limb and head and 6th person to administer medication.
- Limit restraint to a brief period, only to facilitate sedation.
- Apply mechanical restraint (use of a device) only on the instruction of SAPS.

- Remove weapon/mechanism for self-harm.
- Give sedation:
 - Child <13 years: see dose charts.
 - If no IV access obtained (ADULT) give:
 - **midazolam 7.5mg oral OR*
 - *midazolam 10mg buccal/intranasal OR*
 - *diazepam 10mg PO OR*
 - *lorazepam 2mg PO**
 - If IV access obtained (ADULT) give:
 - **midazolam 2mg IV (may repeat IV dose in 15min)**

Give rapid sedation

- Child <13 years: see dose charts
- *midazolam 0.2mg/kg IM (max 15mg) (undiluted). (If >60 years old, give 0.1mg/kg.)*
- If midazolam not available:
 - *lorazepam 4mg IM* OR ^ketamine 4mg/kg IM (for more rapid onset)^

- Conduct a rapid primary survey (once contained) ↔ **PRIMARY SURVEY**
 - If patient cannot maintain own airway, sats <92%, breathing very slow/gasping ↔ **AIRWAY**.
 - If sats <92% give oxygen.
 - Check BP, if <90/60 ↔ **SHOCK ADULT**. (for child see **QUICK REF CHART**)
 - Check glucose: if <3.5mmol/L: smear 25g oral glucose on inside of cheek/gums or, *give 20mL of dextrose 50% IV or if child (<13yrs), give 3mL/kg dextrose 10% IV. If no IV access, give glucagon 1mg IM (child 0.5mg)*. Recheck glucose after 10min. If <3.5mmol/L, repeat.
 - Attempt to obtain IV access once contained.
- Do not assume aggression/agitation is due to a psychiatric condition.
- Look for and manage injury ↔ **HEAD INJURY**; and ask about drug misuse and possible overdose ↔ **POISONING**.

Review patient after 15min of IV meds or 30min of IM/PO meds. Is the patient still aggressive or severely agitated?

Yes: **Administer rapid sedation**

- Child <13 years: see dose charts
 - *Give midazolam 0.1mg/kg IM (max 5mg) (if > 60 years old, give 0.05mg/kg)
 - If midazolam not available: lorazepam 2mg IM*

No

- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP < 92, HR > 100 or < 50, RR > 20 or < 9, sats < 92% or drop in GCS, reassess and manage airway, breathing, circulation. Or age-appropriate vital signs in child ↔ **QUICK REF CHART**.
- Manage underlying conditions ↔ **APPROPRIATE TOOL**.
 - Manage pain ↔ **PAIN**. If airway unsecured, avoid high analgesic doses as these have respiratory depressant effects.
 - If physically restrained or mechanically restrained, check and document that the pulses are all present in each limb.

Document

- **HISTORY** (ask family/patient): history of mental illness, history of any medical illness, epilepsy, medication and any substance misuse or possible poisoning, fever or neck stiffness, history of trauma, especially to the head.
- **EXAMINATION**: responsiveness, seizures, any injuries, vital signs.
- **MANAGEMENT**: medications given, routes, doses and times (as well as response to medications), and procedures performed.



Transport to nearest/most appropriate emergency centre (with family member): you are obligated to transport the patient if any of the following are present and request SAPS instruction for mechanical restraint

- Patient is a danger to him/herself or may be a suicide risk
- Patient is a danger to family/community
- Patient causing serious damage or destruction to property
- SAPS requesting hospitalisation
- Patient has been sedated and needs monitoring

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STROKE

Use this tool for the patient presenting with recent, sudden onset of weakness, slurring of speech or loss of consciousness consistent with a possible stroke.



Do ↔ PRIMARY SURVEY.

Use FAST score to assess for new onset facial drooping, arm weakness and speech difficulty.

F

- **Assess facial movements:** ask patient to smile or show their teeth.
 - Is there an unequal smile or grimace, or obvious facial asymmetry/drooping.

A

- **Assess arm movements:** lift patient's arms together, palms facing up, to 90° if sitting, and 45° if lying supine, and ask them to hold the position for 5 seconds once you let go.
 - Does one arm drift down or fall rapidly?

S

- **Assess for speech impairment:** listen for new onset slurred speech. If available, ask companion. Check for word-finding difficulties: ask patient to name commonplace objects that may be nearby, such as cup, table, keys, pen.
 - Is there a speech impairment?

T

- **Note time of symptom onset.**

Yes to one or more

Facial drooping, arm weakness or speech difficulty present.

Assess and manage urgently as likely stroke:

- Transport rapidly to closest stroke facility, or most appropriate facility if no stroke centre available nearby.
 - Avoid delay to ensure patient is eligible for reperfusion therapy.
- Note which side of body is affected.

No to all

- Look for other causes:
 - Ask about insulin or oral hypoglycaemic medications and check glucose: if $<3.5\text{mmol/L}$, manage hypoglycaemia:
 - Smear 25g glucose on inside of cheek/gum **OR** *give 20mL of dextrose 50% IVI* or if no IV access, give glucagon 1mg IMI*.
 - Repeat glucose every 10min until $>3.5\text{mmol/L}$.
- Ask about history of epilepsy, migraines, possible drug overdose or abuse, or recent trauma and manage accordingly.

- Notify receiving facility of patient and time of onset of symptoms.
- Reassure patient and family members.

Continue to assess and manage ABCDE during transport:

- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP <90/60, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation.
- Manage underlying conditions.
- Document patient's condition including onset, progression, duration as well as all medications given, routes, doses and times and response to medications.



Facial Droop



Arm Weakness



Speech Impairment

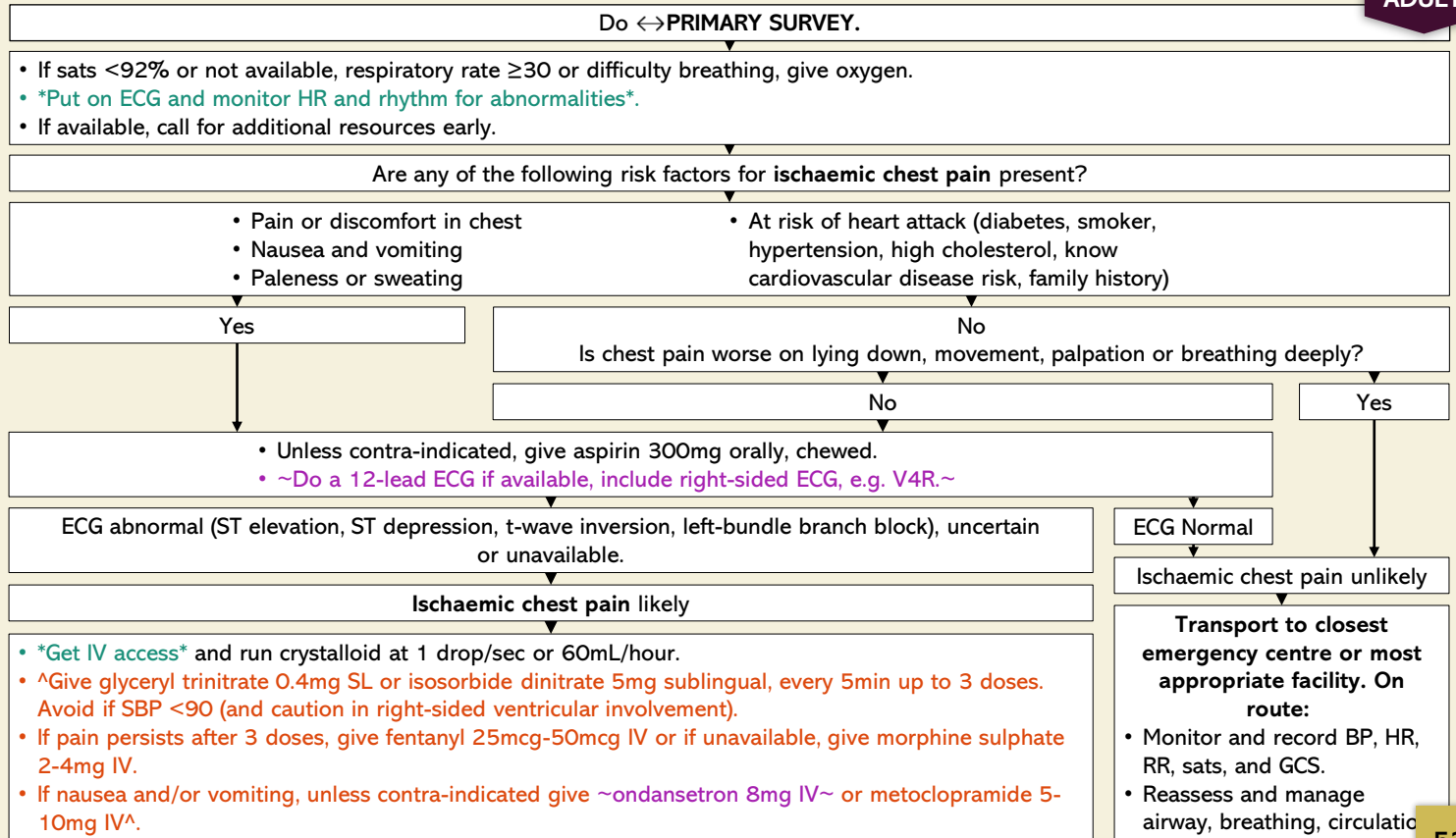


Time of Onset

CHEST PAIN



Use this tool for managing the adult patient presenting primarily with chest pain.



STEMI on ECG and thrombolysis available?

Yes

Check if eligible for prehospital ~thrombolysis if all of the following are present:

- ECG shows: ST-segment elevations ≥ 1 mm in 2 anatomically contiguous leads (except in leads V2 and V3 where the cutoffs are: ≥ 2 mm in man ≥ 40 ys; ≥ 2.5 mm in man < 40 ys, or ≥ 1.5 mm in woman of any age AND
- Onset of symptoms within last 12 hrs AND
- Percutaneous coronary intervention > 20 min away or hospital with thrombolysis > 30 min away.~

Yes

~Does this patient have contraindications to prehospital thrombolysis – any of the following?

- Previous intracranial haemorrhage or stroke; intracranial tumour or arteriovenous malformation.
- Major trauma/surgery/head injury within last 2 weeks; gastrointestinal bleeding in last month or peptic ulcer; known bleeding disorder; suspected aortic dissection (e.g. sharp tearing pain in chest/back or SBP. > 20 mmHg difference between arms); non-compressible punctures in past 24hrs (e.g. liver biopsy, lumbar puncture); active internal bleeding; current warfarin use.
- Hypersensitivity to lytic or if streptokinase given in past year, pregnancy or 1-week post-partum.
- SBP > 180 mmHg and/or DBP > 110 mmHg, ischemic stroke < 3 months ago, CPR.~

No

Eligible for thrombolysis

- ~Consult and send ECG to receiving facility.
- Do prehospital thrombolysis checklist.
- Give clopidogrel 300mg orally give 75mg if ≥ 75 ys.
- Give tenecteplase weight-adjusted dose, AND enoxaparin 30mg IV. Avoid enoxaparin if patient > 75 years **OR**
- If tenecteplase unavailable, give streptokinase 1.5 million IU over 30min~.
- If SBP < 90 slow rate and/or place patient head down.

Yes

Not eligible for thrombolysis or unavailable

Consult with medical advice or receiving facility:

- ~Unless contra-indicated and depending on reperfusion strategy, give clopidogrel 300mg orally.
- Unless contra-indicated and depending on reperfusion strategy, give tirofiban 25mcg/kg bolus IVI followed by 0.15mcg/kg per min infusion.
- With further consultation, start glyceryl trinitrate infusion 5-10mcg/min IV. Avoid if right-side infarction (ST-changes in leads II, III, aVF, or V4R), hypotension (SBP < 90), or use of phosphodiesterase inhibitors (e.g. viagra)~

No

Transport to closest cath lab service, if available, or emergency centre. (Pre-notify the facility of your arrival)

- Monitor and record BP, HR, RR, sats, and GCS every 10min. If BP $< 90/60$, HR > 100 or < 50 , RR > 20 or < 9 , sats $< 92\%$ or drop in GCS, reassess and manage airway, breathing, circulation.

POISONING

Use this tool for the patient where you suspect poisoning – ingestion or contact with a known or unknown harmful substance. ***PLEASE ENSURE THE SCENE AND OWN SAFETY ARE PRIORITISED**

Do ↔**PRIMARY SURVEY**. If convulsing ↔**SEIZURES**.

- **Stabilise ABCDEs:** attach monitors and check sats, HR, BP, RR, and GCS, pupils and CRT:
 - If sats <92%, or RR >30, or RR <9, or GCS ≤14, give oxygen.
 - If airway secretions, suction. If patient unable to maintain airway ↔**AIRWAY**.
 - If breathing inadequate, assist ventilation: give 1 breath every 3-5 seconds using BVM connected to oxygen.

Prevent ongoing exposure

- If poison on patient's skin: protect yourself: wear gloves, apron, eye protection and mask. Remove clothes and wash skin with soap and water.
- If poison in eye/s: immediately flush under running water or normal saline for at least 20min.
- Only if fully conscious and oral overdose of harmful substance in past 1 hour: give activated charcoal 1g/kg mixed with water. Avoid if ingested household products, paraffin, petrol, corrosive poisons (acids/bases), iron, lithium or alcohol. Avoid milk/inducing vomiting.
- If bite/sting observed, expose site and cover with a sterile dressing. Remove any constrictive clothing or jewellery.

Insert IV cannula and assess further according to age:

Adult patient:

- If SBP <90, ***give 250-500mL crystalloid fluid IV over 5-10min.*** Reassess BP: if SBP still <90, repeat bolus. If becomes breathless or new crackles on auscultation, stop IV fluids.
- Check glucose: if <3.5mmol/L, smear 25g glucose on inside of cheek/gum, **OR** give ***20mL dextrose 50% IVI***; if no IV access, give ***glucagon 1mg IMI***. Recheck glucose after 10min. If <3.5mmol/L, repeat.

Child patient (<13yrs): ↔**QUICK REF CHART**

- Assess circulation. If ≥2 of: altered mental status, fast respiratory rate, fast pulse, weak peripheral pulses, delayed cap refill (>2 sec) or low BP (late sign), ***start crystalloid fluid 10mL/kg IV over 10-30min.***
- Check glucose: if <3.5mmol/L, smear 25g glucose on inside of cheek/gum, **OR** ***give 3mL/kg dextrose 10% IVI***; if no IV access, give ***glucagon 0.5mg IMI***. Recheck glucose after 10min. If <3.5mmol/L, repeat.

- **Obtain history from family/friends on scene:** time of exposure, amount and type of substance.
- Ask about all drugs and toxins that may have been accessible: prescribed, over-the-counter, herbal, chemicals, pesticides, traditional medications, recreational substances. Ask to see tablets, packets, containers.
- Check scene for pill bottles, suicide note, fires in poorly ventilated room (cyanide/carbon monoxide poisoning).
- Check for bite/sting/injection marks.

- If poison/substance known, give specific management according to class of drug ↔**SPEC. POISONING**.
- If poison unknown, note unusual smells, look for sweating, drooling, secretions, and check pupils and temperature. Identify familiar toxidromes:

• Agitation, hallucinations and seizures • Sweating • Hyperthermia • Dilated pupils • Tachycardia • Hypertension	• Agitation, hallucinations, seizures, unresponsive • Flushed, dry, warm skin • Hyperthermia • Dilated pupils • Tachycardia • Urinary retention • Decreased bowel sounds • Blurred vision	• Depressed level of consciousness • Hypothermia • Pinpoint pupils • Bradycardia • Hypotension • Respiratory depression	• Depressed level of consciousness, seizures • Sweating • Pinpoint pupils, watery eyes • Bradycardia • Drooling • Urinary incontinence • Diarrhoea and vomiting • Airway secretions/ crackles on auscultation	• Depressed level of consciousness • Slurred speech • Respiratory depression
Sympathomimetic or psycho-stimulant or hallucinogenic, e.g. LSD, cocaine, amphetamines	Anticholinergic, e.g. anti-histamines, atropine	Opioid, e.g. morphine, codeine, methadone, heroine, fentanyl, tramadol	Cholinergic, e.g. organophosphates, carbamates	Sedative-hypnotic, e.g. alcohol, benzodiazepines
↔ SPEC. POISONING p2	↔ SPEC. POISONING p2	↔ SPEC. POISONING p1 ↓	↔ SPEC. POISONING p1	↔ SPEC. POISONING p2

- Wherever possible, contact Poisons Information Helpline of the Western Cape (0861 555 777) for specific management recommendations.
- Check if toxic dose taken: use total mg ingested ÷ weight, to get mg/kg. Check package insert/discuss to determine if toxic for that substance.
 - If toxic dose or patient symptomatic with no response to treatment or poisoning due to unknown substance, transport to closest emergency centre. If possible and safe to do so, take remaining tablets/packets/containers with to the emergency centre.
 - If non-toxic dose and patient is well, with no signs or symptoms, advise patient to remain at home and monitor for changes.

Monitoring during transportation

- If initially SBP < 90 or signs of shock in a child, reassess to see if further fluids needed ↔**SHOCK IN ADULT, CRITICALLY ILL CHILD**.
- Monitor BP, HR, RR, sats, and GCS every 10min. If glucose was low initially, recheck.
 - If BP <90/60, HR >100 or <50, RR >20 or <9, sats <92% or drop in GCS, reassess and manage airway, breathing, circulation, level of consciousness. Or age-appropriate vital signs in child ↔**QUICK REF CHART**.
 - If unable to maintain own airway, sats <92, breathing very slow/gasping →**AIRWAY**.

Specific poisoning management page 1

	ADULT management	CHILD management
Opioid poisoning	- If respiratory depression: - Give *naloxone 0.4mg IV/IM*. If known opioid abuser, give *naloxone 0.04mg IV/IM* to limit withdrawal symptoms. - Reassess every 2min: - If respiratory rate still low, give increasing doses of *naloxone: 0.8mg, 2mg, 4mg every 2min, up to a maximum total dose of 10mg*. - If respiratory rate remains low, consider other poisoning.	- If respiratory depression: - Give *naloxone 0.1mg/kg IV/IM, up to 2mg. - Reassess every 2min: - If respiratory rate still low, give repeat dose of *naloxone 0.1mg/kg every 2min, up to a maximum cumulative dose of 10mg*. - If respiratory rate remains low, consider other poisoning.
	Note: naloxone wears off quickly, monitor respiratory rate closely and if needed, give further doses later.	
Cholinergic poisoning (organo-phosphate)	- Give ^atropine 2mg IV^. - Look for a response after 5min - clear lung sounds, dilated pupils, increased HR, dry mouth: - If no response: ^double the previous atropine dose every 3min until clinical improvement^. - If response: Calculate the total amount of atropine given as boluses. ^Give 20% of this dose per hour via IV infusion^. - If unable to maintain airway or signs of toxicity persist after initial management, consider intubation as per ↔AIRWAY. - If RSI, avoid succinylcholine as induction agent.	- Give ^atropine 0.05mg/kg IV^. - Reassess after 5min: - If no response - clear lung sounds, dilated pupils, increased HR, dry mouth: ^double the previous atropine dose every 3min until clinical improvement^. ^When patient responds to atropine. Calculate the total amount of atropine given as boluses. Give 20% of this dose per hour^. - If unable to maintain airway or signs of toxicity persist after initial management, consider intubation as per ↔AIRWAY. - If RSI, avoid succinylcholine as an induction agent.
Tricyclic antidepressant poisoning	- Attach ECG if not already done - If signs of toxicity: QRS widened, ventricular arrhythmia, seizures or hypotension not responsive to fluids: - Give ^sodium bicarbonate 8.4% 100mL IV over 3min^. Reassess after 10min: - If signs of toxicity persist, repeat dose. - If signs of toxicity persist after this, consult senior medical advisor/Poisons Information Helpline for further guidance. - If unable to maintain airway or signs of toxicity persist after initial management, consider intubation as per ↔AIRWAY. ^If intubated, hyperventilate with 100% oxygen at a rate of 20-24 breaths per minute^. - Aim for ETCO2 target of 20-25mmHg.	- Attach ECG if not already done, if QRS widened, ventricular arrhythmia, seizures or hypotension not responsive to fluids: - Give ^sodium bicarbonate 8.4% 2mL/kg IV over 3min^. Reassess after 10min - If signs of toxicity persist, repeat dose. - If signs of toxicity persist after this, consult senior medical advisor/Poisons Information Helpline for further guidance. - If unable to maintain airway or signs of toxicity persist after initial management, consider intubation as per ↔AIRWAY.

Specific poisoning management page 2

	ADULT management	CHILD management
Beta-blocker and/or calcium-channel blocker poisoning	- If bradycardic ↔ RCSA BRADYCARDIA - If hypotensive, give 250-500mL crystalloid fluid IV bolus over 5-10min. - If still hypotensive, give [^]calcium chloride 10mL 10% IVI over 10mins[^] OR [^]calcium gluconate 30mL 10% IVI over 10min[^] under ECG monitoring. - If still hypotensive, give [^]adrenaline (1mg in 50mL of normal saline or 4mg in 200mL normal saline)[^]. - Give via peripheral line, not with other drugs and use infusion pump or rate control device. - Start at 5mL/hour (=1.6mcg/min) with target dose of adrenaline 6-30mL/hour (2-10mcg/min). - Titrate to effect by increasing by 1mL/hour every 5min until BP and pulse increase – keep rate constant when reaches BP 90 systolic (MAP 65).	- If bradycardic ↔ RCSA BRADYCARDIA - If hypotensive, give 10mL/kg crystalloid fluid bolus over 5-10min. - If still hypotensive, give [^]calcium chloride 0.2mL/kg 10% IVI, OR calcium gluconate 0.6mL/kg 10% IVI, administered at 1mL/min[^] under ECG monitoring. - If hypotensive and beta-blocker poisoning more likely, give [*]glucagon 100mcg/kg IVI over 1-2min (not exceeding 1mg/minute)[*]. If inadequate effect after 10min, repeat dose. - If still hypotensive, give [^]adrenaline (0.06mg/kg in 50mL of normal saline or 5% dextrose)[^]. - Give via peripheral line, not with other drugs and use infusion pump or rate control device. - Start at 1mL/hour with target dose of adrenaline 0.02-0.1mcg/kg/min. - Titrate to effect by increasing by 1mL/hour every 5min until signs of shock improve.
Sedative-hypnotic poisoning	- If temperature ≥ 38°C: remove clothing and use water spray and fan or ambulance air-conditioning to cool patient. - If aggressive or agitated, ↔ AGGRESSION/AGITATION - If unable to maintain airway, consider ↔ AIRWAY. - If cocaine or other stimulant overdose suspected: - If chest pain from suspected stimulant use: - Give aspirin 300mg orally. - Give [^]diazepam 5mg or midazolam 1mg IVI slowly[^]. - Give [^]sublingual isosorbide dinitrate 5mg or sublingual glyceryl trinitrate 400mcg via metered dose spray[^]. - Exclude cardiac cause ↔ CHEST PAIN. - Attach ECG if not already done, if QRS widened: give [^]sodium bicarbonate 8.4%[^] as above.	- If temperature ≥ 38°C: remove clothing and use water spray and fan or ambulance air-conditioning to cool patient. - If aggressive or agitated, ↔ AGGRESSION/AGITATION. - If unable to maintain airway, consider ↔ AIRWAY. - Attach ECG if not already done, if QRS widened: give [^]sodium bicarbonate 8.4%[^] as above.
Anticholinergic poisoning		
Sympathomimetic or psychostimulant poisoning		

PATIENT IN PAIN



Use this tool to assess and adequately manage pain according to patient's response. An unconscious patient may still feel pain.

Do ↔ PRIMARY SURVEY.

- Communicate with patient and family: recognise and acknowledge fear.
- Reassure that their condition is treatable, and pain will be managed with their input, so they will be able to control pain relief.
- Assess and document patient's pain
 - If unconscious, look for signs of pain and manage likely degree of pain according to presentation: facial expressions (tearing, frowning, grimacing), restless (withdrawing from pain), guarding, tensing or stiff. Check vital signs: increased HR, SBP, RR may indicate pain.
 - If conscious, assess degree of pain using scales appropriate for age as below.

Child less than 4 years old (can be used up to 7 years)

Look at facial expression, position of legs, activity level, type of cry and ability to be consoled. Score each of these using the table below and add scores to get a total to rate pain:

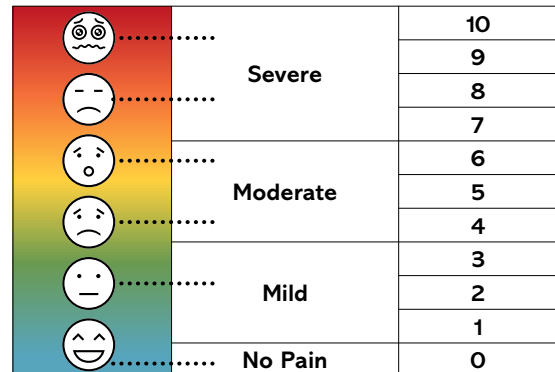
Score	0	1	2
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, uninterested	Frequent to constant quivering chin, clenched jaw
Legs	Normal position or relaxed	Uneasy, restless, tense	Kicking, or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, shifting, back and forth, tense	Arched, rigid or jerking
Cry	No cry (awake or asleep)	Moans or whimpers; occasional complaint	Crying steadily, screams or sobs, frequent complaints
Ability to be consoled	Content, relaxed	Reassured by occasional touching, hugging or being talked to, distractible	Difficult to console or comfort

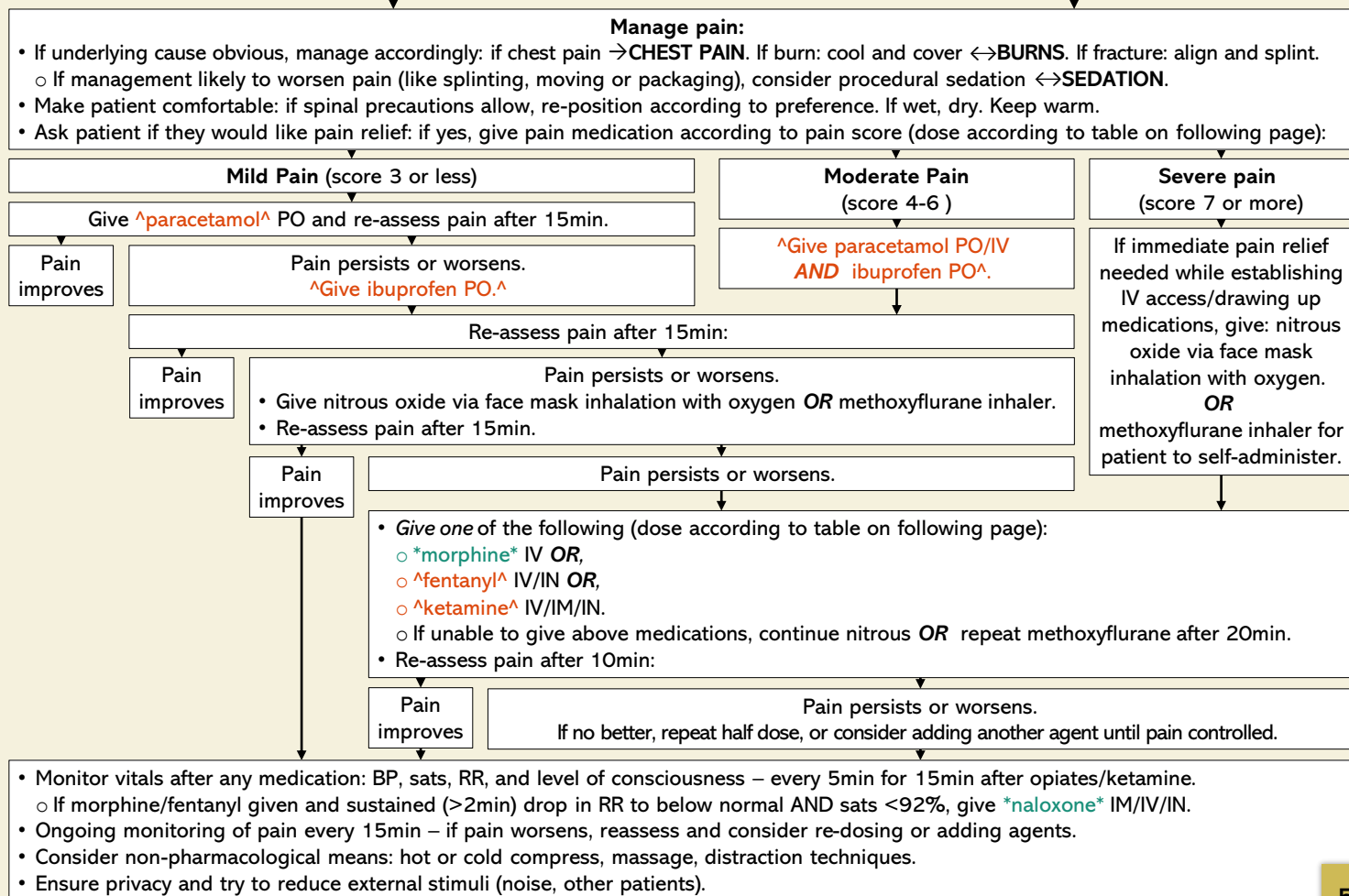
Child 4-13 years old

Ask child to choose a face that shows how bad their pain is right now.

Adult (≥13 years)

Ask to rate pain from 0-10: zero being no pain, ten being worst pain they can imagine. If unable, ask if pain is mild, moderate or severe.





Pain	Pain Medication	Dosing for adults or child ≥13 years	Dosing for child <13 years	Notes (possible side-effects and additional key notes)
Mild	[^] Paracetamol [^]	PO: 1g (give up to 4g in 24hrs) IV: 1g over 15min	PO: 15mg/kg (up to 75mg/kg in 24hrs) – see dosing paracetamol dosing table ↔ DRUG APPENDIX . IV: 15mg/kg 6hrly	Avoid if severe liver problem.
Moderate	[^] Ibuprofen [^]	PO: 400mg (max 2.4g in 24hrs)	PO: 5-10mg/kg (max 40mg/kg/24hrs) see ibuprofen dosing table ↔ DRUG APPENDIX .	Avoid if pregnant (especially in third trimester), severe kidney or liver problems or active stomach ulcer disease.
	Nitrous oxide	Inhaled: give 50:50 (nitrous oxide:oxygen) via face mask with demand valve – allow patient to self-administer.		Avoid if head injury, decreased GCS, hypotension (SBP <90), sats <92%.
	Methoxyflurane (Penthrox/ Pentthoxyflurane)	Inhaled: give 3mL inhaler- instruct patient to inhale and exhale gently through the mouthpiece. Allow patient to self-administer over 30min. May be repeated after 20min, up to total dose of 6mL.		Avoid in decreased GCS, respiratory depression, hypotension (SBP <90mmHg), prior anaesthetic reaction.
Severe	[^] Fentanyl [^]	IV: 50mcg IN: 100mcg	IV: 1mcg/kg IN: 1.5mcg/kg	• Rapid onset, short duration. • Inject slowly and titrate to effect. • If no better, repeat ½ dose after 5min. • Avoid if respiratory depression.
	Morphine	IV: 2.5mg injected slowly every 5min, titrated to effect, usually up to 10mg (consider max of 15mg) – usual dosing range of 0.05mg/kg – 0.1mg/kg	IV: 0.1mg/kg	• Slower onset and longer duration. • Avoid if severe liver problem or respiratory depression. • May cause hypotension – ensure adequate initial BP, give slowly and with fluids and monitor carefully.
	[^] Ketamine [^]	Analgesic doses: IV: 0.1-0.3mg/kg, usually 10mg. IM: 0.5mg/kg, usually 25mg. IN: 1mg/kg, usually 50mg.	IV: 0.25-0.5mg/kg IM: 0.5-1mg/kg IN: 0.5-1mg/kg	• Rapid onset of dissociation, duration moderate. • Avoid if eclampsia, coronary disease (disease of arteries supplying heart), or stroke. • Strongly consider this as a co-agent. • Caution in catecholamine depleted hypotension
	Naloxone (for reversal of opioid overdose)	IV/IN/IM: 0.4mg given slowly titrated to effect (max 2mg). Repeat dose every 2-3min until effect.	IV/IN/IM: 0.1mg/kg max 2mg. Repeat dose every 2-3min until effect.	• Use if sustained respiratory depression due to morphine/fentanyl (or other opiates/ drugs of abuse). • Short duration of action (45min) - continue to monitor closely as further doses of naloxone may be needed during transport.

PROCEDURAL SEDATION

Use this tool if managing the patient is likely to worsen pain (like splinting, moving or packaging).

- Anticipate increased pain from procedures such as splinting, moving, extrication or packaging. If needed, sedate before painful procedure.
- Except for ketamine, sedation does not provide analgesia – give both unless brief, non-painful procedure.
- If able, explain process to patient before. Give patient permission to say stop or hold up hand to say stop if in pain.
- Ensure oxygen, suction, BVM, and IV access available and working.
- Attach monitors and monitor vitals: ECG, BP, RR, sats (and ETCO₂ if available).
- **Avoid administering procedural sedation to children under 2 years old**, except under senior medical guidance.

Use one or more of the below sedative agents. If protracted pain anticipated, give in combination with pain relief medications.

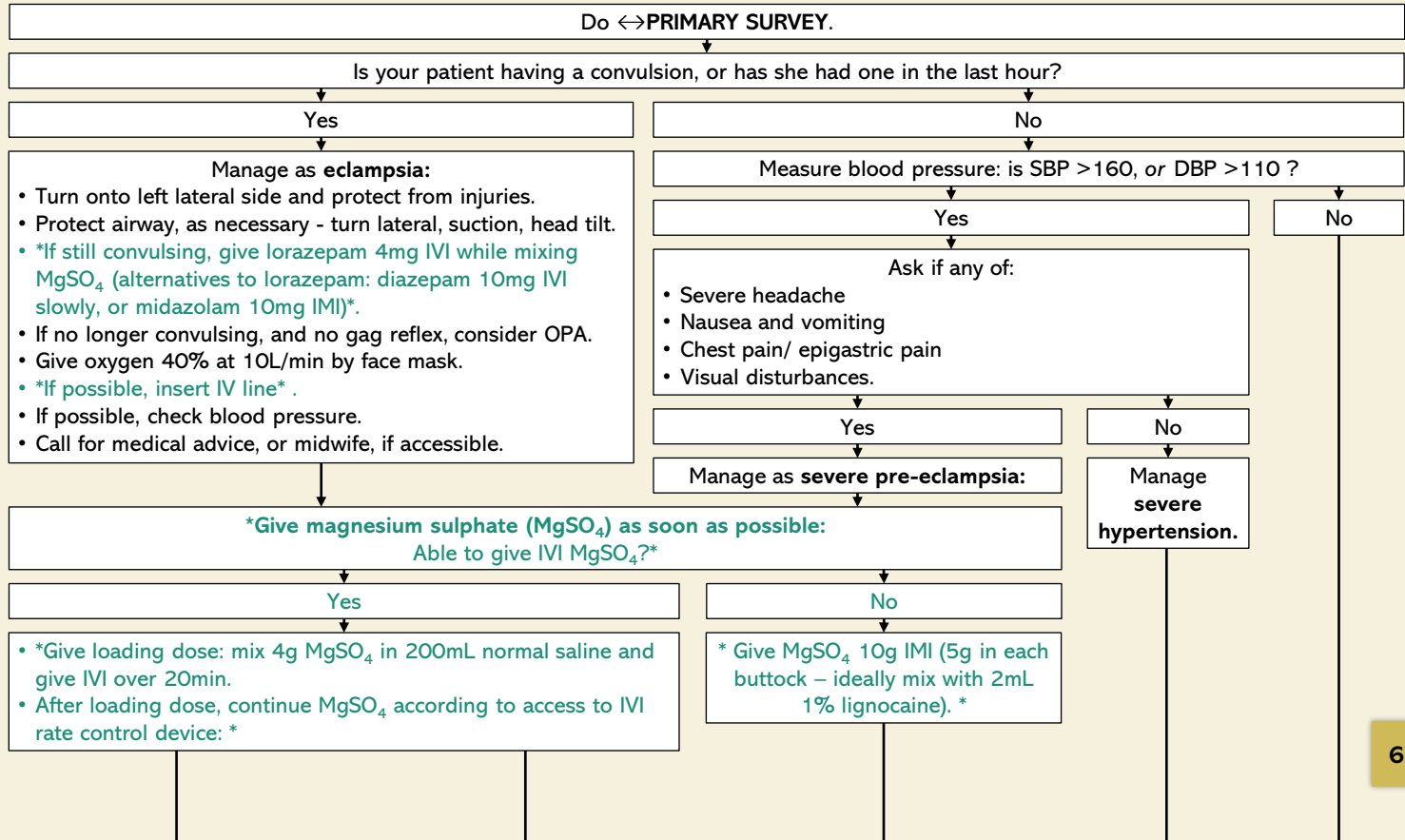
Sedation Medication	Dose	Notes
[^] Ketamine [^]	Sedation doses: IM: 4-5mg/kg IV/IN: 1-2mg/kg	Rapid onset of dissociation, duration moderate. Administer slowly and titrate to effect. Duration 15-30min. Can awake confused and combative.
~Etomidate~	IV: 0.2-0.4mg/kg	Rapid, deep and short duration sedation. Administer slowly. Beware of transient hypoventilation.
Midazolam	IV: 0.1mg/kg	Unpredictable. Slow onset of action can take 5min, do not repeat dose before 5min. Duration 5-10min.
Methoxyflurane (Pentrox/ pentthoxyflurane)	Inhaled: 3-6mL titrated to effect	For rapid and not very painful procedures.
<i>Use the agents below primarily for pain – not sedation.</i>		
[^] Fentanyl [^]	IV: 2mcg/kg	Analgesic effect greater than sedative. Rapid onset, can cause transient hypoventilation and hypotension.
Morphine	IV: 0.1mg/kg	Analgesic effect much greater than sedative. Longer effect than fentanyl, can also cause transient. hypoventilation and hypotension: consider preload with IV fluids to ensure adequate fluid status.

HYPERTENSION AND/OR CONVULSIONS IN PREGNANCY



Use this tool for any woman presenting with high blood pressure and/or convulsions in pregnancy (i.e. after 20 weeks' gestation (i.e. uterus at level of umbilicus or higher), or soon after delivery).

Ask family if patient is known with epilepsy: if known with epilepsy manage as epileptic convulsion ↔ SEIZURES.



*** Access to IVI rate control device:**

Mix MgSO_4 4g in 200mL normal saline and give at a rate of 1g/hour (50mL/hr) *

*** No rate control device**

Give MgSO_4 10g IMI (5g in each buttock – ideally mix with 2mL 1% lignocaine). *

~Control blood pressure

- Repeat BP: if SBP still ≥ 160 or DBP still ≥ 110 , give nifedipine 10mg PO to swallow. Repeat BP after 30min:
 - If SBP still ≥ 160 or DBP still ≥ 110 : repeat nifedipine 10mg PO.
 - Aim for target BP of: SBP < 140 and DBP < 90 .

If convulsions recur or persist after loading dose MgSO_4 , give additional 2g MgSO_4 in 200mL normal saline IVI over 10min.

Transport

- Monitor vitals every 15min (BP, HR, RR, sats, GCS):
 - If BP $< 90/60$, HR > 110 or < 50 , RR > 24 or < 9 , sats $< 92\%$ or drop in GCS, reassess and manage airway, breathing, circulation, level of consciousness. Ideally, paramedic or experienced nurse to accompany patient.
- Restrict IV fluids: TKVO only.
- Transport in lateral position with head slightly raised.
- If possible, document in maternity case record observations, timing and MgSO_4 administration.
- Expedite transport and decide on transfer destination - ensure receiving facility has specialist obstetric and neonatal care capabilities.

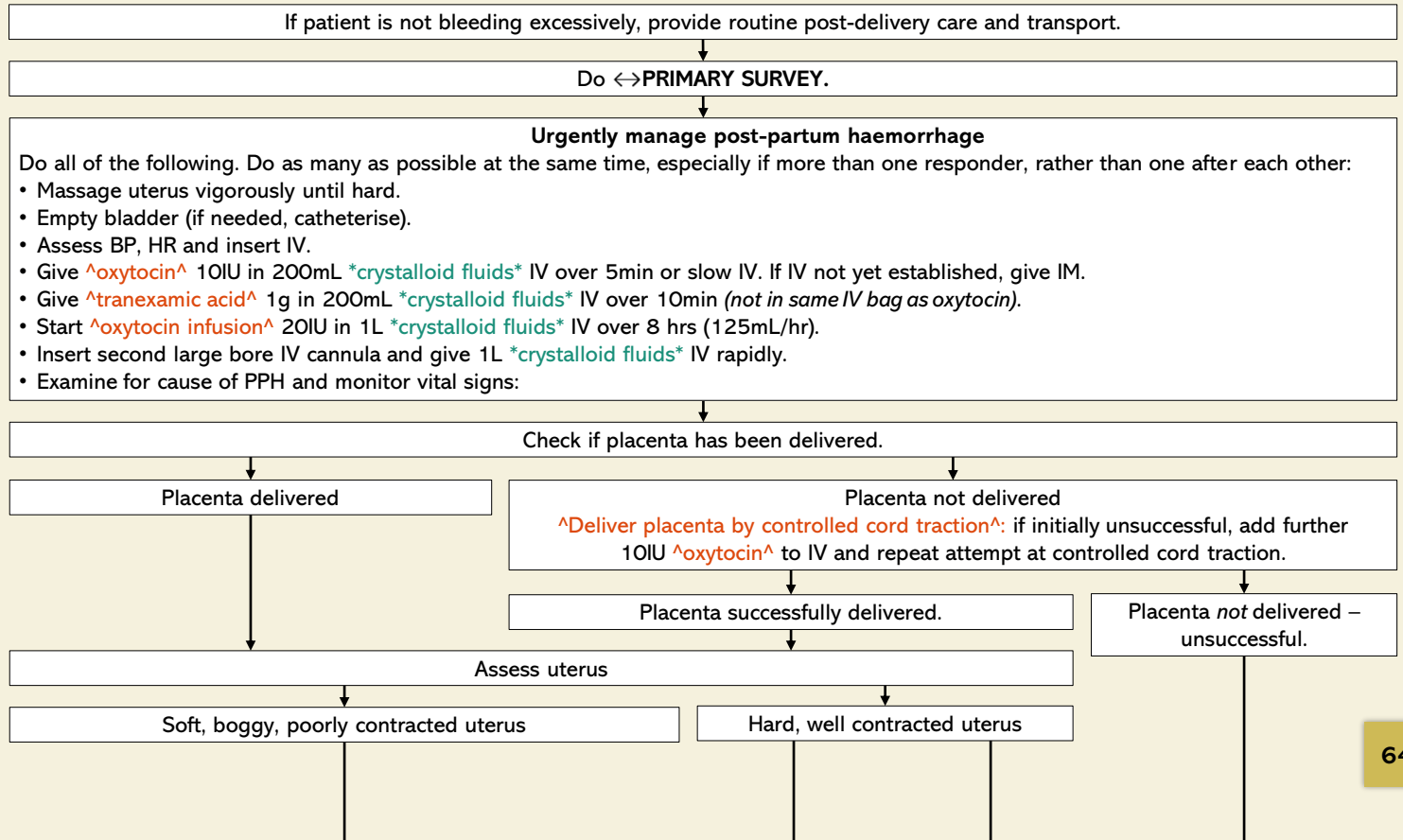
Consider baby

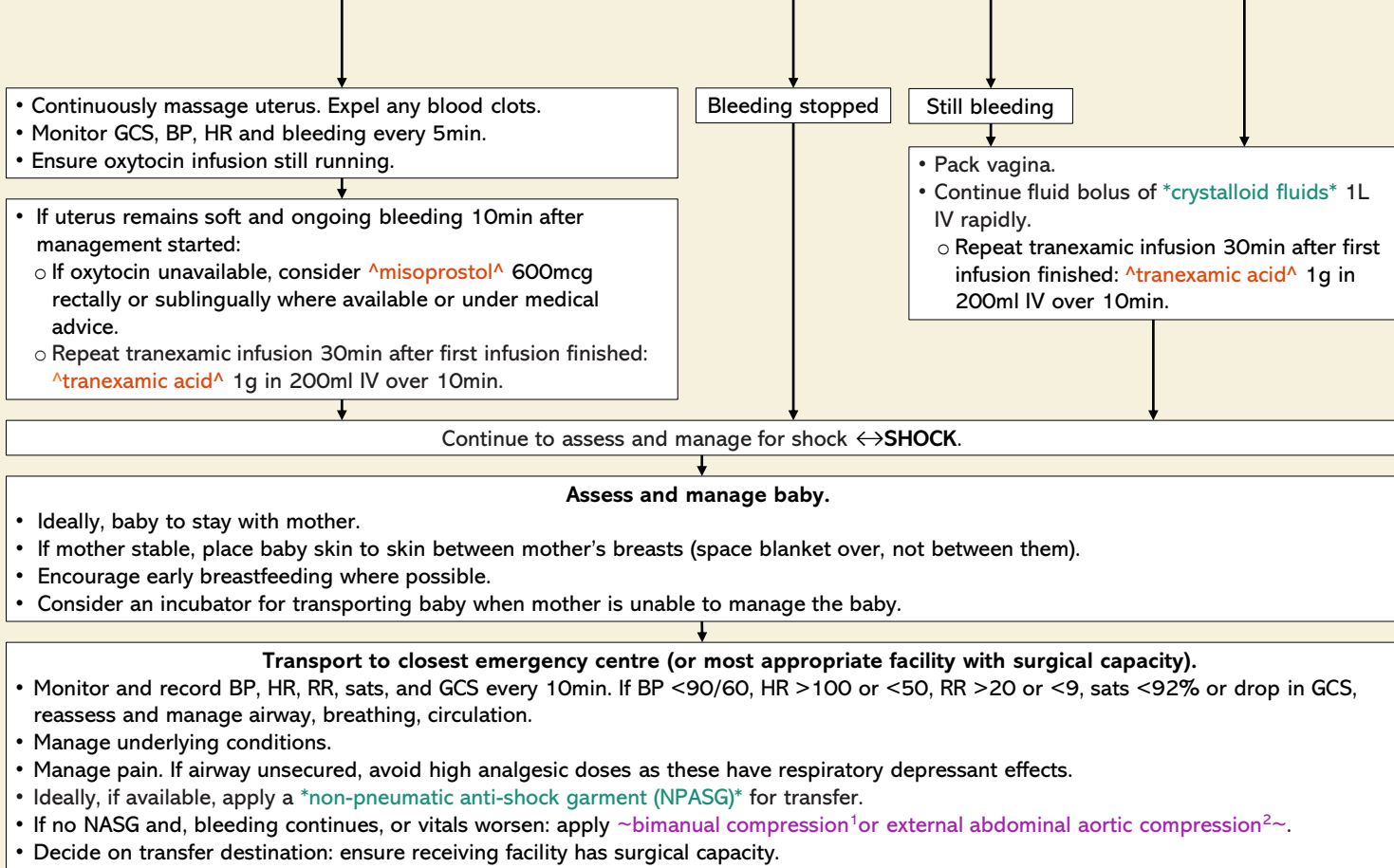
- If already delivered, assess and manage baby:
 - Ideally, baby to stay with mother.
 - If mother stable, place baby skin to skin between mother's breasts with space blanket over, not between them.

VAGINAL BLEEDING AFTER CHILDBIRTH



Use the algorithm below to manage the patient who presents with active vaginal bleeding after childbirth that is considered excessive, either by measurement (>500mL), or by clinical judgement (pad/cloth soaked in <5min or constant trickling of blood). This is referred to as post-partum haemorrhage.





¹Bimanual compression of the uterus: place one hand over uterus externally; place other hand in vagina to apply pressure on lower segment of uterus. Compress consistently with the 2 hands to compress uterus to reduce blood flow.

²External abdominal aortic compression is performed by applying firm and sustained pressure to the aorta above the level of the umbilicus while transferring to hospital.

CARING FOR THE ADULT AT THE END OF LIFE



Use this tool for the patient known to be at the end of their life who is deteriorating despite treatment

If unsure if patient considered terminal, the following may indicate end-of-life care in place:

- On oral morphine.
- Fully dependent on a carer for all needs of daily living.
- 'Do not resuscitate (DNR)' orders, a living will, or an advance care plan or other legal documentation.
- Presence of common end-of-life signs and symptoms:
 - Cachexia (loss of body mass), profound weakness, bedbound
 - Unconscious, comatose, disoriented
 - Difficulty swallowing
 - Not eating or drinking



Obtain history and review medications

- Ask about the main concern that prompted the call out (like pain or difficulty breathing). Ask about other symptoms.
- Review medications: ask when last medications were given. If a dose has been skipped, assist family to give the medication according to prescription. If home oxygen available and not in yet in use, start oxygen.
- Ask about who the main caregiver is at home and talk about their concerns.



Share the decision making

- Ask about goals of care, and what is most important – is there specific treatment the patient/family does or does not want.
- Ask about preference to stay home or be transported to hospital.
- Consult with patient's palliative care doctor/nurse, or regular general practitioner. If unavailable, consult with medical command.
- Be respectful, non-judgmental and sensitive.



Avoid procedures or interventions that may cause discomfort

- Avoid IV insertion and suctioning, unless immediately necessary.
- If terminal phase e.g. decreased consciousness, unable to take swallow, CPR is inappropriate: prepare family – explain that it will not restore function for any significant period, that it prolongs the dying process and is distressing for the family.

Assess and manage symptoms

- Assess for pain, noisy/difficulty breathing, pain, restlessness/anxiety/agitation and nausea/vomiting and manage accordingly:

Pain

- Position patient comfortably.
- If on oral morphine:
 - If missed analgesia dose, give now.
 - If no missed dose, and patient still in pain, **^give half the prescribed dose of oral morphine stat^**.
- If not yet on morphine, give **^morphine 1-2mg SC^**.

Noisy breathing

- If noisy breathing, excessive secretions likely: try changing position. Position on side and make comfortable with pillow to allow fluids to drain from mouth.
- This is distressing for family: reassure that this is common and not causing patient discomfort.
- Avoid suctioning – it is ineffective in long run and is distressing.

Difficulty breathing

- Position patient comfortably.
- Ensure cool airflow across patient's face (with a fan where possible).
- If not already on oxygen, start it.
- If wheezing, give ***salbutamol 5mg and ipratropium bromide 0.5mg via inhaler or nebuliser***.
- Give **^morphine 1-2mg SC^**.

Restlessness/ anxiety/agitation

- Look for and manage reversible causes: ask about pain, full bladder, last bowel motion, nausea, and fears and anxiety.
- Once underlying causes addressed, give **^midazolam 5mg SC^**.
- Minimise sensory stimulation.

Nausea/vomiting

- Try changing position. If able, assist patient to sit upright.
- Minimise sights and smells such as from food and cooking.
- Advise family that cold food, with smaller portions may minimise nausea and vomiting.
- Give **^metaclopramide 10mg SC^**.

Manage further according transport decision

- In consultation with next of kin, and/or medical oversight, meet patient wishes regarding transport to medical facility.
- If patient wishes to remain at home:
 - Link with local healthcare workers/palliative care teams. Suggest accessing psychosocial/spiritual care support.
 - Emphasise the importance of giving pain medication regularly and daily use of a laxative to prevent constipation.
 - Emphasise prevention of pressure ulcers: advise to wash and dry skin daily. Keep linen dry. Move (lift, avoid dragging) patient every 1-2 hours if unable to shift own weight. Look daily for skin colour changes and seek care for these early.
- Document the transport decision clearly, and hand patient over to next of kin.
- Explain what to do when the patient does die: caregivers should contact a medical provider to declare patient dead, then contact funeral services to collect the mortal remains.

DRUG APPENDIX (TO BE READ WITH APPROVED LIST OF MEDICATIONS PER REGISTRATION CATEGORY)

Medication	Scope (to be read with approved list of medications from PBEC)	Indication (which CDST)	Route	Dosing for Adults or child ≥ 13 years	Dosing for Child < 13 years	Notes (including important information for administration, contra-indications, things to watch for and extremes of ages)
Activated charcoal	ALL	Poisoning	PO/NG	50-100g	1g/kg - See paediatric dosing tables	Mix with water. Avoid if ingested household products, paraffin, petrol, corrosive poisons (acids/bases), iron, lithium, or alcohol.
Adrenaline	AEA ECA ECT ANT ECP	Airway	IV	5-20mcg bolus		Mix 1mg of 1:1000 adrenaline into 200mL sodium chloride 0.9% (5mcg/mL solution) and draw 4mL into syringe. Give 1mL of mixture IV at a time, titrating to effect.
		DIB adult	IM	0.5mg		Concentration 1:1000
		DIB child	NEB		0.5mL/kg of 1:1000 (max 5mL)	
		Poisoning/ Shock Adult	IV infusion	2-10mcg/min (start 5mL/hr)	0.02-0.1mcg/kg/min (start at 1mL/hr)	- Adult: mix 1mg in 50mL of normal saline or 4mg in 200mL normal saline; target dose of adrenaline in range 6-30mL/hr. - Child: mix 0.06mg/kg in 50mL of normal saline or 5% dextrose.
Atropine	ECA ECT ANT ECP	Airway	IV		0.02mg/kg	Infant < 3 months old: min 0.1mg, max 0.4mg
		Poisoning	IV	2mg	0.05mg/kg - See paediatric dosing tables	
Calcium chloride	ANT ECP	Poisoning	IV	1g 10% over 10min	0.2mL/kg 10% at 1mL/min	
Calcium gluconate	ANT ECP	Poisoning	IV	3g 10% over 10min	0.6mL/kg 10% at 1mL/min	
Clopidogrel	ANT ECP	Chest Pain	PO	300mg		If ≥ 75 yrs: give 75mg.

Medication	Scope (to be read with approved list of medications from PBEC)	Indication (which CDST)	Route	Dosing for Adults or child >13 years	Dosing for Child ≤ 13 years	Notes (including important information for administration, contra-indications, things to watch for and extremes of ages)
Dextrose	AEA ECA ECT ANT ECP	ALL	IV	20mL of 50% (consider dilution – add 20ml of sterile water/0.9% Normal Saline)	• 3mL/kg of 10% • See paediatric dosing tables	*Neonate not for admin by ECA.
Diazepam	ECA ECT ANT ECP	Seizures	IV	10mg	• 0.25mg/kg • See paediatric dosing tables	Neonate: 0.3-0.4mg/kg
		Seizures	PR	10mg	• 0.5mg/kg (max 10mg) • See paediatric dosing tables	Neonate: 1.25-2.5mg/kg
		Aggression	PO	10mg	0.2mg/kg (max 10mg/dose)	
Enoxaparin	ECP	Chest Pain	IV	30mg		Avoid enoxaparin if patient >75 years.
Etomidate	ECP	Airway	IV	0.3mg/kg	0.3mg/kg	Give slowly
Fentanyl	ANT ECP	Pain	IN	100mcg	1.5mcg/kg	
		Pain	IV	50mcg	1mcg/kg	
		Airway	IV	1-3mcg/kg	1-3mcg/kg	
		Chest Pain/ Poisoning	IV	25-50mcg		
Furosemide	ANT ECP	DIB adult	IV	40mg		
Glucagon	AEA ECA ECT ANT ECP	ALL	IM	1mg	0.5mg	For child >20kg ± >6 yrs age: consider 1g dose according to some sources.
	ECT ANT ECP	Poisoning	IV		100mcg/kg over 1-2min	Max 1mg/min; max 10mg/dose.
Glucose (oral)	ALL	ALL	BUCCAL	25g	<5ys 5g; 5-11yrs 10g; >12yrs 15g	Smear inside cheek/gums.

Medication	Scope (to be read with approved list of medications from PBEC)	Indication (which CDST)	Route	Dosing for Adults or child >13 years	Dosing for Child ≤ 13 years	Notes (including important information for administration, contra-indications, things to watch for and extremes of ages)
Glyceryl trinitrate	ECT ANT ECP	Chest Pain	SL	0.4mg		50g in 50mL of normal saline. Increase every minute until clinical effect.
Hydrocortisone	AEA ECA ECT ANT ECP	DIB adult/ child	IV	200mg	5mg/kg See paediatric dosing tables	
Ibuprofen	ANT ECP	Pain	PO	400mg (max 2.4g in 24hrs)	• 5-10mg/kg • See paediatric dosing tables	
Ipratropium bromide	ALL	DIB adult	NEB	0.5mg		
Isosorbide dinitrate	ECT ANT ECP	DIB adult/ Chest Pain	SL	5mg		
			IV	1-2mg/hour		10mg in 50mL of normal saline (usually 10ml of drug + 40mL of normal saline). Increase until clinical effect (max 8 – 10mg/hour)
Ketamine	ANT ECP	Pain	IM	0.5mg/kg, usually 25mg	0.5-1mg/kg	
		Pain	IN	1mg/kg, usually 50mg	0.5-1mg/kg	
		Pain	IV	0.1-0.3mg/kg, usually 10mg.	0.25-0.5mg/kg	
		Procedural Sedation/ Aggression	IM	4-5mg/kg	4-5mg/kg	
		Procedural Sedation	IV/IN	1-2mg/kg	1-2mg/kg	
		Airway	IV	2mg/kg	1-2mg/kg	
		Airway/ Ventilation	IV infusion	0.25-0.5mg/kg/hr	1-2mg/kg/hr	See Airway Management Tool.
Lorazepam	ECA ECT ANT ECP	Seizures	BUCCAL		0.1mg/kg	
		Seizures/ Aggression	IM	4mg	0.1mg/kg	
		Seizures	IV	2mg	0.1mg/kg	Neonate: 0.1mg/kg
		Aggression	PO	2mg	0.1mg/kg (max 4mg)	

Medication	Scope (to be read with approved list of medications from PBEC)	Indication (which CDST)	Route	Dosing for Adults or child >13 years	Dosing for Child ≤ 13 years	Notes (including important information for administration, contra-indications, things to watch for and extremes of ages)
Magnesium sulphate	AEA ECA ECT ANT ECP	Eclampsia	IM	10g		5g in each buttock – ideally, mix each 5g with 1mL 1% lignocaine.
	ECT ANT ECP	Eclampsia	IV	4g		Mix in 200mL saline and give IVI over 20min (1g/hour (50mL/hr).
		Eclampsia	IV infusion	1g/hour (50mL/hr)		Following loading dose, mix 4g in 200mL normal saline.
	ECT ANT ECP	DIB adult	IV	1-2g		Mix in 200mL sodium chloride 0.9% IV over 20min.
Methoxyflurane (Penthoxyflurane)	ALL	Pain	INH	3mL	3mL	Inhale - self-administer over 30min. May be repeated after 20min, up to total dose of 6mL.
Metoclopramide	ECT ANT ECP	Chest Pain	IV	5-10mg		
		End of Life	SC	10mg		

Medication	Scope (to be read with approved list of medications from PBEC)	Indication (which CDST)	Route	Dosing for Adults or child >13 years	Dosing for Child ≤ 13 years	Notes (including important information for administration, contra-indications, things to watch for and extremes of ages)
Midazolam	ECA ECT ANT ECP	Seizures	IN		0.2mg/kg	
		Seizures	BUCCAL	10mg	• 0.5mg/kg • See paediatric dosing tables	Neonate: 0.3mg/kg
		Seizures	IM	10mg	0.15mg/kg	
		Seizures	IV	10mg	• 0.2mg/kg • See paediatric dosing tables	Neonate: 0.15mg/kg
		Aggression	BUC/ IN	10mg	0.2mg/kg (max 10mg/ dose)	
		Aggression	IM	0.2mg/kg	0.15mg/kg (max total dose 10mg)	Max 15mg adult, >60yrs 0.1mg/kg.
		Aggression	IV	2mg	0.05mg/kg (max total 0.5mg/kg)	
		Aggression	PO	7.5mg	0.5mg/kg (max total 20mg)	
		Procedural Sedation	IV	0.1mg/kg	0.1mg/kg	
		End of Life	SC	5mg		
		Airway/ Ventilation	IV infusion	0.06-0.24mg/kg/hr	0.05-0.2mg/kg/hr	See Airway Management Tool.
Misoprostol*	Medical advisement	PPH	PR/SL	600mcg		*Not currently in scope.
Morphine	ECT ANT ECP	Pain/ Chest Pain	IV	2.5mg (max 15mg)	0.1mg/kg	Repeat dose every 2-3min until effect.
		End of Life	PO	Related to prescribed dose		If on oral morphine: • Missed analgesia dose, in pain: give full prescribed dose. • No missed analgesia dose, still in pain: give half of the prescribed dose.
		End of Life	SC	1-2mg		
		Airway/ Ventilation	IV infusion	0.02-0.08mg/kg/hr	0.02-0.08mg/kg/hr	See Airway Management Tool.
Naloxone	AEA ECA ECT ANT ECP	Poisoning/ Pain (Opioid Reversal)	IV/ IM/ IN	0.4mg (max 2mg)	0.1mg/kg (max 2mg)	Repeat dose every 2-3min until effect.

Medication	Scope (to be read with approved list of medications from PBEC)	Indication (which CDST)	Route	Dosing for Adults or child >13 years	Dosing for Child ≤ 13 years	Notes (including important information for administration, contra-indications, things to watch for and extremes of ages)
Nifedipine	ECP	Eclampsia	PO	10mg		
Nitrous oxide	ALL	Pain	INH	Self admin	self admin	Give via face mask with demand valve.
Ondansetron	ECP	Chest Pain	IV	8mg		
Oxytocin	ECT ANT ECP	PPH	IV	10U in 200mL over 5min		Or 10U slow IVI.
		PPH	IM	10U		
		PPH	IV infusion	20U/8hrs (125 mL/hr)		Mix 20IU in 1L crystalloid fluid.
Paracetamol	ANT ECP	Pain/ Crit Ill Child	IV	1g over 15min	15mg/kg 6 hourly	
		Pain	PO	500mg (max 4g in 24hrs)	15mg/kg (max 75mg/kg in 24hrs) – See paediatric dosing tables.	
Rocuronium	ECP	Airway/Ventilation	IV	1.2mg/kg	1.2mg/kg	
Salbutamol	ALL	DIB adult/ child	INH/ NEB	5mg	- MDI 10 puffs - NEB <3yrs: 0.5mL - NEB > 3yrs: 1mL	MDI: Use spacer <3yrs. Nebulisation – mix with 4mL normal saline.
Sodium bicarbonate	ANT ECP	Poisoning	IV	100mL 8.4% over 3min	2mL/kg 8.4% over 3min	
Streptokinase	ECP	Chest Pain	IV	1.5 million IU over 30min		
Succinylcholine	ECP	Airway	IV	2mg/kg	1-2mg/kg	
Tenecteplase	ECP	Chest Pain	IV	Weight adjusted dose		<60kg: 30mg; 60-70kg: 35mg; 70-80kg: 40mg; 80-90kg: 45mg; >90kg: 50mg
Thiamine	AEA ECA ECT ANT ECP	ALL	IV/IM	100mg		Given in hypoglycaemia when concern of vitamin deficiency (e.g. malnourishment or history of chronic alcohol use).
Tirofiban (GPIIb/IIIa Inhibitor)	ECP	Chest Pain	IV	25mcg/kg bolus		Followed by 0.15mcg/kg per min infusion.
Tranexamic acid	ANT ECP	PPH	IV	1g in 200mL over 10min		
	ANT ECP	Badly Inj	IV	2g over 10min		

FORMULARY

If child's age and weight not in same row, choose dose according to weight.

Activated charcoal		
Oral, 1g/kg (up to 100g) mixed as a slurry with water		
Weight (kg)	Dose	Age
3.5-7kg	5g	1-6 months
7-11kg	10g	6-18 months
11-17kg	15g	18 months – 5 years
17.5-35kg	25g	5-11 years
35-55kg	50g	11-15 years
≥55kg	100g	≥ 15 years

Atropine				
IV, 0.05mg/kg/dose (up to 3mg)				
Weight (kg)	Dose (mg)	Use one of the following injections		Age
		0.5mg/mL	1mg/mL	
3.5-5kg	0.2mg	0.4mL	0.2mL	1-3 months
5-7kg	0.3mg	0.6mL	0.3mL	3-6 months
7-9kg	0.4mg	0.8mL	0.4mL	6-12 months
9-11kg	0.5mg	1mL	0.5mL	12-18 months
11-14kg	0.6mg	1.2mL	0.6mL	18 months – 3 years
14-17.5kg	0.8mg	1.6mL	0.8mL	3-5 years
≥ 17.5kg	1mg	2mL	1mL	≥ 5 years

Adrenaline			
IM/IV, 1 mg/mL (1:1 000), 0.01mL/kg as a single dose			
Weight (kg)	Dose (mg)	Injection 1mg/mL (1:1 000)	Age
< 9kg	0.05mg	0.05mL	< 1 year
9-12kg	0.1mg	0.1mL	1-2 years
12-17.5kg	0.2mg	0.2mL	2-5 years
17.5-40kg	0.3mg	0.3mL	5-12 years
≥ 40kg	0.5mg	0.5mL	≥ 12 years
≥ 55kg	1mg	1mL	≥ 15 years

Diazepam			
Rectal, 0.5mg/kg/dose for fits as a single dose			
Weight (kg)	Dose (mg)	Ampoule 10mg/2mL	Age
3-6kg	2mg	0.4mL	< 6 months
6-10kg	2.5mg	0.5mL	6 months – 1 year
10-18kg	5mg	1mL	1-5 years
18-25kg	7.5mg	1.5mL	5-8 years
≥25kg	10mg	2mL	≥ 8 years

Diazepam	
IV, 0,25mg/kg/dose (up to 5mg if < 5 years old and 10mg 5 years old) for fits as a single dose	
Weight (kg)	Ampoule 10mg/2mL (=5mg/mL)
4-5kg	0.2mL
5-6kg	0.25mL
6-7kg	0.3mL
7-8kg	0.35mL
8-9kg	0.4mL
9-10kg	0.45mL
10-11kg	0.5mL
11-12kg	0.55mL
12-13kg	0.6mL
13-14kg	0.65mL
14-15kg	0.7mL
15-16kg	0.75mL
16-17kg	0.8mL
17-18kg	0.85mL
18-19kg	0.9mL
19-20kg	0.95mL
20-25kg	1.0mL
≥25kg	0.25 x weight ÷ 5 = number of mL to give

Hydrocortisone			
Slow IV, 4-6mg/kg (up to 100mg) immediately			
Weight (kg)	Dose (mg)	Injection 100mg/2mL	Age
11-14kg	50mg	1mL	2-3 years
14-17.5kg	75mg	1.5mL	3-5 years
≥ 17.5kg	100mg	2mL	≥ 5 years

Ibuprofen				
Oral, 5-10mg/kg/dose (up to 400mg) 8 hourly with food				
Weight (kg)	Dose (mg)	Use one of the following:		Age
		Syrup	Tablet	
		100mg/5mL	200mg	
9-11kg	80mg	4mL		12-18 months
11-14kg	100mg	5mL		18 months – 3 years
14-17.5kg	120mg	6mL		3-5 years
17.5-25kg	150mg	7.5mL		5-7 years
25-40kg	200mg	10mL	1 tablet	7-12 years
≥ 40kg	400mg		2 tablets	≥ 12 years

Midazolam			
- Buccal (between the cheek and gum), 0.5mg/kg (up to 10mg) - Check formulation: use 5mg/mL formulation for buccal administration.			
Weight (kg)	Dose (mg)	Buccal 5mg/mL	Age
< 4kg	2mg	0.4mL	< 2 months
4-7kg	3mg	0.6mL	2-6 months
7-9kg	4mg	0.8mL	6-12 months
9-11kg	5mg	1mL	12-18 months
11-14kg	6mg	1.2mL	18 months – 3 years
14-17.5kg	7.5mg	1.5mL	3-5 years
≥ 17.5kg	10mg	2mL	≥ 5 years

Paracetamol				
Oral, 10-15mg/kg/dose (up to 1g) 6 hourly, as needed for up to 5 days.				
- If < 1 month: maximum daily dose is 60mg/kg/day. - If ≥ 1 month: maximum daily dose is 90mg/kg/day (up to 4g daily).				
Weight (kg)	Dose (mg)	Use one of the following:		Age
		Syrup	Tablet	
		120mg/5mL	500mg	
3.5-5kg	48mg	2mL		1-3 months
5-7kg	72mg	3mL		3-6 months
7-9kg	96mg	4mL		6-12 months
9-11kg	120mg	5mL		12-18 months
11-14kg	144mg	6mL		18 months – 3 years
14-17.5kg	180mg	7.5mL		3-5 years
17.5-25kg	240mg	10mL	½ tablet	5-7 years
25-35kg	360mg	15mL		7-11 years
35-55kg	500mg		1 tablet	11-15 years
≥ 55kg	1000mg		2 tablets	≥ 15 years

Midazolam		
IV, 0.25mg/kg (up to 10mg)		
Weight (kg)	Midazolam 5mg/5mL ampoule (=1mg/mL)	Midazolam 15mg/3mL ampoule (=5mg/mL)
< 4kg		0.2mL
4-5kg	1mL	0.3mL
5-6kg	1.3mL	
6-7kg	1.5mL	
7-8kg	1.8mL	0.4mL
8-9kg	2mL	
9-10kg	2.3mL	0.5mL
10-11kg	2.5mL	
11-12kg	2.8mL	0.6mL
12-13kg	3mL	
13-14kg	3.3mL	0.7mL
14-15kg	3.5mL	
15-16kg	3.8mL	0.8mL
16-17kg	4mL	
17-18kg	4.3mL	0.9mL
18-19kg	4.5mL	
19-20kg	4.8mL	1mL
20-25kg	5mL	
25-35kg	7.5mL	1.5mL
≥ 35kg	9mL	1.8mL

Formulary sourced from PACK Child, 2024.

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DEXTROSE 10% FOR PAEDIATRIC HYPOGLYCAEMIA

How to mix a 10% dextrose solution



Small Child <10 kg (<1 years)

Add 4mL 50% dextrose to 16mL normal saline or Ringers Lactate to make 20mL of 10% dextrose:

- Take 20mL syringe.
- Draw up 16mL normal saline or Ringers Lactate.
- Then draw up 4mL of 50% dextrose.
- Mix well.

Older Child >10kg (>1years)

Add 40mL 50% dextrose to 160mL normal saline or Ringers Lactate to make 200mL of 10% dextrose:

- Open 200mL bag of normal saline or Ringers Lactate.
- Remove and discard 40mL of fluid from this bag – leaving 160mL in the bag.
- Next, draw up and inject 40mL of 50% dextrose into the bag.
- Mix well.

Give by slow IV injection 3mL/kg of 10% Dextrose
(Followed by a maintenance infusion of dextrose containing fluid)

Weight (kg)	Dose (mL of 10% dextrose)	Age
2-4kg	10mL	0 - 1 month
4-6kg	15mL	1- 6 months
6-10kg	25mL	6 months – 1 year
10-18kg	40mL	1-5 years
18-25kg	65mL	5-8 years
25-35kg	80mL	8-11
>35kg	100mL	>11 years

QUICK REFERENCE CHART

Decide if pulse rate is normal for age		
Age	Pulse rate (beats/minute)	
	Pulse rate decreased if:	Pulse rate increased if:
< 1 year	< 110	≥ 160
1-2 years	<100	≥ 150
2-5 years	< 95	≥ 140
5-12 years	< 80	≥ 120
≥ 12 years	< 60	≥ 100

How to estimate weight using age*	
- If < 12 months old: (Age in months +9) / 2 - If 1-5 years old: (Age +4) x 2 - If > 5 years old: Age x 4 <i>*Length based tools such as the Pawper or Broselow tapes give the most accurate weight estimation where available.</i>	
Age	Weight (kg)
<2 months	<4kg
2-6 months	4-7kg
6-12 months	7-9kg
12-18 months	9-11kg
18 months – 3 years	11-14kg
3-5 years	14-17.5kg
5-7 years	17.5 - 25kg
7-11 years	25-35kg
11- 15 years	35-55kg
>15 years	>55kg

Decide if respiratory rate is normal for age		
Age	Respiratory rate (breaths/minute)	
	Respiratory rate decreased if:	Respiratory rate increased if:
0-2 months	< 30	≥ 60
2-12 months	< 30	≥ 50
1-5 years	< 25	≥ 40
5-12 years	< 20	≥ 25
≥ 12 years	< 15	≥ 20

- If rapid, simple assessment of LOC needed, use AVPU scale.
- If more detailed assessment of LOC needed (like head injury or considering intubation), use GCS.

Assess level of consciousness (LOC) using the Glasgow Coma Scale (GCS)

- Add scores to give a single score out of 15.

Best motor response

- Spontaneous movements (obeys commands)
- Withdraws to touch (localizes pain)
- Withdraws from pain
- Abnormal flexion to pain
- Extends to pain
- None

Best verbal response

- Age-appropriate vocalization, smile, or response to sound; interacts (coos, babbles); follows objects
- Cries, irritable
- Cries to pain
- Moans to pain
- None

Decide if blood pressure is normal for age		
Age	Systolic blood pressure (mmHg)	
	Blood pressure decreased if:	Blood pressure increased if:
< 1 year	< 80	> 90
1-2 years	< 85	> 95
2-5 years	< 85	> 100
5-12 years	< 90	> 110
≥ 12 years	< 100	> 120

Assess level of consciousness with AVPU

- A** Alert
- V** responds to Voice
- P** responds to Pain
- U** Unresponsive/Unconscious

Eye opening

- Spontaneous
- To voice
- To pain
- None



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